

STUDY GUIDE · TEST AND EXAM PREPARATION

Introduction to Engineering

Weeks 1–4. Every savollar, topshiriq, atama, formula and qoida from the thirteen course files — lectures, labs, the syllabus and three textbooks — with model answers, a glossary for every week and a cheat-sheet of key facts.

Introduction to Engineering · CC-INEN010104 · 4 ECTS

Tashkent University of Technology · Fall 2026

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SAVOLLAR · TOPSHIRIQLAR · ATAMALAR · FORMULALAR ·
QOIDALAR

Contents

1. Course map, sources and how to use this guide	3
1.1. How the course and the grading system work	4
2. Week 1 — Engineering as a Profession	5
2.1. Key points (the «all qoidalar (rules)» of Week 1)	5
2.2. Week-1 ethics qoidalar (rules): the NSPE codes and personal ethics	8
2.3. Week-1 savollar (questions) — lecture and lab questions with model answers	10
2.4. Savollar (questions) and topshiriqlar (assignments) from the textbooks for week 1 (with solutions) . . .	12
2.5. Atamalar (terms) — Week-1 glossary	16
3. Week 2 — Systems Thinking I: systems, boundaries, stakeholders	17
3.1. Key points (“all qoidalar” of week 2)	17
3.2. Week-2 savollar (questions) — lecture and lab questions with model answers	19
3.3. Week-2 savollar (questions) and qoidalar (rules) from the textbooks	21
3.4. Week-2 atamalar (terms) — glossary	25
4. Week 3 — Systems Thinking II: process flow diagrams (PFD)	26
4.1. Key points (“all qoidalar” of week 3)	26
4.2. Week-3 savollar (questions) — questions with model answers	29
4.3. Week-3 savollar (questions) and formulas from the textbooks: mass balance (Kosky ch. 18)	31
4.4. Week-3 atamalar (terms) — glossary	32
5. Week 4 — Problem Framing and Modelling I	33
5.1. Key points (“all qoidalar” of week 4)	33
5.2. Week-4 savollar (questions) — questions with model answers	39
5.3. Week-4 savollar (questions) and qoidalar (rules) from the textbooks	42
5.4. Week-4 atamalar (terms) — glossary	49

6. Formulalar (Formulas) — consolidated reference of all course formulas	51
7. Qoidalar (Rules) — every course rule in one checklist	53
7.1. Process and diagram rules	53
7.2. Problem statement and variable rules	54
7.3. Rules of units, significant figures, and calculations	54
7.4. Ethical rules	55
7.5. Design and team rules	55
7.6. Rules of the course project milestones	56
8. Topshiriqlar (Assignments) — all course assignments, weeks 1–4	56
9. Cheat sheet: key facts, numbers and quotes	57
9.1. Facts and numbers they ask about	57
9.2. Quotes that embellish an answer	58
9.3. Ten one-line answers (for quick revision)	59
9.4. What comes next in the syllabus (weeks 5–8, a preview)	59

1. Course map, sources and how to use this guide

This study guide has been compiled by hand from all 13 course files of «Introduction to Engineering» (code CC-INEN010104, 4 ECTS, fall semester 2026/27, Tashkent University of Technology, Faculty of Power Engineering). It covers the material of the first four weeks: every savol (savollar — questions), topshiriq (topshiriqlar — assignments), atama (atamalar — terms), formula (formulalar) and qoida (qoidalar — rules) that appears in the lectures, the laboratory sessions, the syllabus and the three textbooks is written out in full and supplied with a model answer. The guide is built so that you can prepare for the weekly mini projects (30%), the midterm (30%) and the final (40%) without going back through hundreds of slides and pages.

The course is taught by Dr. Mohammad Reza Bahrami (lectures; Dean of the Faculty of Power Engineering, PhD in Mechanical Engineering, professor at Innopolis University, adjunct professor at Constructor University Bremen) and Ibromkhimjon Khamidov (labs; assistant professor, MSc/PhD track at Politecnico di Milano, research — lean manufacturing, sustainability, quantitative productivity assessment). The course follows the Project-Based Learning approach: all semester work revolves around one team system, which each team chooses in week 2 and carries through to the final report.

Wk	Week topic (per syllabus)	Lecture	Lab	Textbook chapters
1	Engineering as a profession: culture, identity, historical, social and economic roles; how engineering differs from science and design	Lecture01 — Engineering as a Profession (33 slides)	Lab 1 — Engineering as a Profession (17 slides)	Kosky ch. 1; Moaveni ch. 1–2
2	Systems thinking I: systems as interacting elements, processes and stakeholders; system boundaries and the operational context	ITE-F26-Lecture2 — Systems Thinking I (21 slides)	Lab 2 — Systems Thinking I (15 slides)	Kosky ch. 2 (variables); Moaveni ch. 6
3	Systems thinking II: process flow diagrams (PFD); how different perspectives lead to different readings of the same system	ITE-F26-Lecture2 — Systems Thinking II (29 slides)	Lab 3 — Systems Thinking PFD (14 slides)	Kosky ch. 18 (mass balance, Example 18.2)
4	Problem framing and modelling I: turning informal observations into a well-defined engineering problem; identifying variables	Lecture04 — Problem Framing and Modelling I (33 slides)	Lab 4 — Problem Framing I (16 slides)	Kosky ch. 21–23; Dym ch. 1, 3–6; Moaveni ch. 3, 6

Table 1.1. Weeks 1–4 and the course files on which this guide is based

The three textbooks the lectures build on: (1) P. Kosky, R. Balmer, W. Keat, G. Wise, «Exploring Engineering: An Introduction to Engineering and Design», 5th edition, Academic Press, 2020 — the course's main text, listed in the syllabus as recommended reading; all four lectures refer to it. (2) S. Moaveni, «Engineering Fundamentals: An Introduction to Engineering» — the source on the profession, ethics, design and units of measurement. (3) C. L. Dym, «Engineering Design: A Project-Based Introduction» — the source on problem framing: project goals, constraints and functions. The lectures cite Kosky's chapters explicitly: chapter 2 (engineering variables, units, significant figures, the Need-Know-How-Solve method),

chapter 18 (Example 18.2 — mass balance), chapter 21 (the eight steps of the design process), chapter 23 (the Empire State Building, blind spots, Blimp Wars, D/W requirements).

1.1. How the course and the grading system work

From the syllabus: the workload is 120 hours (4 ECTS $\times$ 30 h) — 30 hours of lectures, 30 hours of in-class team project work, 60 hours of independent study. There are no prerequisites, and the course is open to all first-year students; what is expected is working with basic quantitative logic, active participation in teamwork, and clear oral and written communication in English. The five intended learning outcomes (ILOs), with their Bloom's taxonomy levels and mapping to the ASIIN competencies:

- **ILO-1 (Understand, subject competency)** — explain that engineering models are purposeful simplifications of complex systems, and describe how assumptions and approximations shape a model's usefulness and limitations.
- **ILO-2 (Apply, subject competency)** — apply basic quantitative reasoning (demand and capacity, rates and flows, order-of-magnitude estimates) to characterize the behavior of a real system from incomplete data.
- **ILO-3 (Analyze, methodological competency)** — analyze system dynamics (delays, queues, capacity, bottlenecks) and use simple simulations under uncertainty.
- **ILO-4 (Evaluate, systemic competency)** — evaluate trade-offs between competing objectives (efficiency, cost, fairness, accessibility) and justify a design decision taking technical, ethical and stakeholder considerations into account.
- **ILO-5 (Apply, social competency)** — collaborate effectively in a team throughout the semester project and present models, assumptions and decisions clearly in writing and orally.

Component	Format	Duration	Weight	ILO
Weekly mini projects	Report + presentation	30 minutes	30%	1–4
Midterm Exam	Written exam on paper	1–2 hours	30%	1–4
Final Exam	Written exam on paper	1–2 hours	40%	6

Table 1.2. Course assessment (section 9 of the syllabus and the Assessment Methods and Completion screenshot)

Grading scale: 90–100 points = A (excellent); 75–89 = B (good); 60–74 = C (satisfactory); 0–59 = D (fail).

Course qoidalar (rules), from the syllabus. Attendance: to be admitted to the final exam you must attend at least 75% of all classes; missing 25% or more means you lose the right to sit the final. Plagiarism: zero tolerance — a first violation gives 0 for the assignment (a second attempt is possible at the instructor's discretion), a second gives 0 with no resubmission, and a third is an automatic fail of the course. AI tools (ChatGPT, Claude, DeepSeek and the like) are allowed for clarifying concepts and debugging, but any AI-generated content counts as a source: you must state the tool explicitly and document how it helped; submitting AI text as your own work without the instructor's permission is forbidden — punished under the plagiarism rules.

How to use this guide: Sections 2–5 follow the week-by-week chronology; within each week comes first the digest of «all the week's qoidalar (rules)» (the material that is checked first of all), then all the savollar (questions) of the lectures and labs with model answers, then the questions and assignments copied out of the textbooks for that week's topic, and at the end a glossary of the week's atamalar (terms). Sections 6–9 are round-ups: all the formulas, all the assignments, all the rules, and a cheat sheet of key facts. Terminology is kept bilingual (English + Uzbek), because the exam is conducted in English.

2. Week 1 — Engineering as a Profession

2.1. Key points (the «all qoidalar (rules)» of Week 1)

Etymology and definition. The word «engineer» comes from the Latin ingenium — «the fruit of genius, a talented invention» — and the verb ingeniare — «to devise, to invent». About 2,000 years ago ingenium described the designing of a new machine; the word then shifted to the machines themselves, and later to the people who designed them — the «engine-ers». The same root survives today in the French/German ingénieur and the Italian ingegnere. Course definition: an engineer is «a creative, inventive person who finds solutions to practical problems». The modern definition of engineering: engineering is the practice of using natural science, mathematics and the engineering design process to solve technical problems, to raise efficiency and productivity, and to improve systems. Kosky (ch. 1) adds: engineers use creative design and analysis — involving energy, materials, motion and information — to serve human needs, and they express their knowledge in the form of variables, numbers and units.

«Science can amuse and fascinate us all, but it is engineering that changes the world» — Isaac Asimov.

«Scientists investigate that which already is; engineers create that which has never been» — Albert Einstein.

Fields of engineering. The classical fields from the lecture: civil (roads, bridges, water systems), mechanical (machines, engines, robots), electrical (power, circuits, control), computer (hardware and software), chemical (processes, materials, fuels), aerospace (aircraft and spacecraft), biomedical (medical devices), environmental (water, air, sustainable development); plus agricultural, ceramic, industrial, manufacturing, marine, materials, mining, nuclear, petroleum, textile and others. A historical note from Kosky: the oldest kind of engineering is military (fortifications, siege works, roads for armies); the first non-military engineers

were called civilian engineers, which was shortened to civil engineers; civil engineering is one of the oldest continuously practiced disciplines.

Engineering as a Profession: Classic Disciplines

Week 1 · the eight disciplines from the lecture and their focus

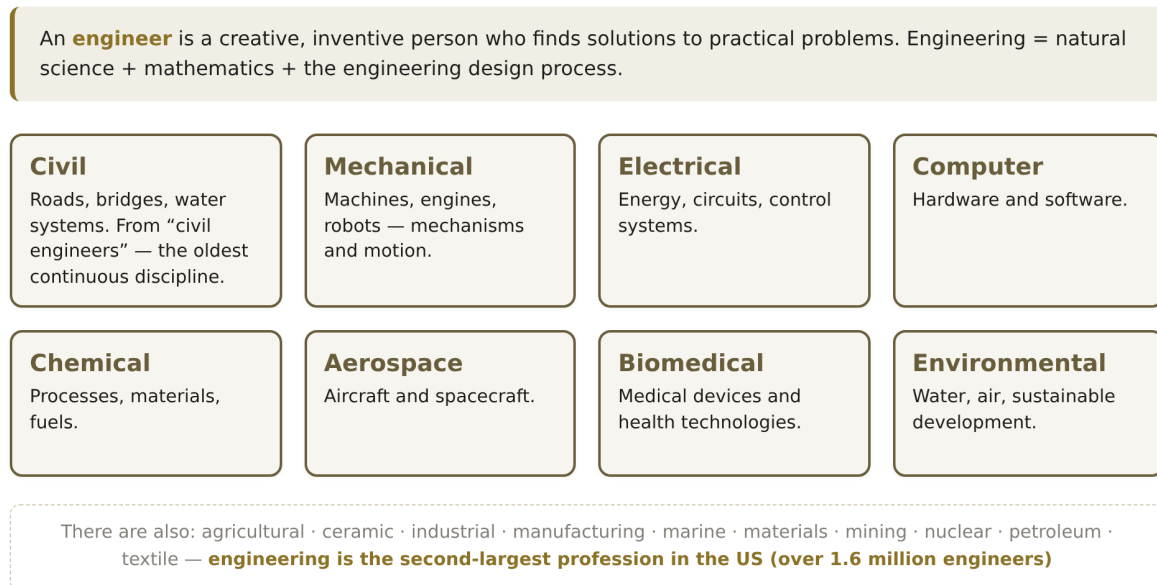


Fig. 1. Classical fields of engineering (week 1): the eight disciplines from the lecture and their focus.

A common misconception (NOT THE SAME AS). Engineering is not repair. A plumber installs and fixes pipes — an engineer designs the water supply system: pipe diameters, pressures and layouts are chosen before anything is built. A car mechanic repairs a vehicle — an engineer designs its parts: the engine, the transmission and the safety systems that the mechanic later services. This distinction is a favorite comprehension-check question: engineering = design and analysis before, and instead of, repair.

Historical role. The four eras from the lecture: the Industrial Revolution (steam, railways, factory production), electrification (electric power and telecommunications connecting cities and industries), the age of aviation and the automobile (flight and mass mobility, compressing distance and time), and the digital and information age (computing, the internet, software). In each era it was engineers who turned new scientific knowledge into the tools, systems and infrastructure that people actually use. A quote from James Kip Finch: «The engineer has been, and is, a maker of history».

Social role. Engineers shape the everyday conditions of life: the water we drink, the energy in our homes, hospitals, roads. Engineers sit «at the interface between science and society» (Gordon Stanley Brown, MIT): they translate scientific possibilities into products, systems and infrastructure that reach millions of people, so their decisions carry direct social consequences — safety, health, fairness, the environment. That is exactly why engineering is organized as a profession, with codes of ethics and standards of practice. The mission of the century was formulated by the US National Academy of Engineering (NAE Grand Challenges — 14 in total, 6 are given in the lecture): provide access to clean water; restore and improve urban infrastructure; advance health informatics; make solar energy economical; secure cyberspace; engineer

better medicines (the full list also covers energy, the brain, learning and scientific discovery).

Economic role (the lecture's numbers): more than 1.6 million engineers work in the United States alone; engineering is the country's second-largest profession; the average starting salary of an engineering graduate in the US is \$68,000 a year (2017 data from NACE, repeated in Kosky ch. 1). An engineering degree is a springboard: graduates routinely move into medicine, law, business and public service. Beyond careers, engineering is an engine of national development: this is how countries build the infrastructure, industry and technological base on which economic growth depends. Moaveni (ch. 1, U.S. Bureau of Labor Statistics data) gives the employment distribution: 1,572,100 engineers in total, of them civil — 278,400 (17.7%), mechanical — 238,700 (15.2%), industrial — 214,800 (13.7%), electrical — 157,800 (10.0%); most engineering degrees are awarded in the electrical, mechanical and civil specializations — the «parents of all the other branches».

Culture, identity and professional societies. The word «engineer» is strikingly uniform across languages — a reflection of a shared identity that crosses borders. Engineers usually work in teams — from two or three people to hundreds on a single large project (for example, the design of a passenger aircraft). Professional societies reinforce that shared identity: they define the standards, continuing education and ethics of each discipline. The key societies of the lecture: ASME — mechanical; IEEE — electrical/computers; ASCE — civil; AIChE — chemical; AIAA — aerospace; ANS — nuclear. Kosky adds ASTM (materials and testing) and BMES (biomedical). A typical engineering society (a Kosky footnote) defines the disciplines required for membership, advocates for them, sets codes and standards, provides additional educational courses and maintains its own code of ethics.

The ten traits of the «problem-solver DNA» (mindset of a good problem solver) — the shared list of the lecture and Lab 1: objective, analytical, creative, dedicated, curious, courageous, adventurous, enthusiastic, patient, vigilant. In the lab, each person chooses their trait, and groups are formed so that the mix of traits is diverse: «a group of five Objective members will argue with itself forever and never solve anything». Moaveni (ch. 1) gives a parallel list of the common traits of good engineers: engineers are problem solvers; they have a firm command of fundamental principles; they are analytical, attentive to detail and creative; they want to learn throughout their lives; they carry a portable core of knowledge (a good mechanical engineer can work as an automotive, aerospace or chemical engineer); they have written and oral communication skills; they manage their time; they have «people skills»; they write reports; they use computers freely for modeling; they participate actively in professional organizations; they work in a team environment.

Science — engineering — design (the week's triad). Science investigates what already exists, asks «why does this happen?» and looks for explanations, generalizable laws and truth about the natural world. Engineering creates what has never existed, asks «how do we make it work?» and applies scientific knowledge under real constraints to build a solution. Design is the creative, open-ended act of giving a solution its form: shape, aesthetics and user experience; engineering design is design disciplined by analysis: physics, materials, cost, manufacturability and safety constrain the possibilities. The «bridge test»: if a bridge is given a striking shape with no regard for load-bearing capacity, that is design without engineering; if the shape is derived from an analysis of loads, materials and structural integrity, that is engineering design. Steve Jobs's quote from the lecture: «Design is not how it looks like and feels like. Design is how it works». The connection

scheme: science understands the natural world — design shapes creative solutions — engineering builds working solutions; engineering draws on science and expresses itself through design.

2.2. Week-1 ethics qoidalar (rules): the NSPE codes and personal ethics

Ethics is the central theme of week 1 (the lecture + all of Lab 1 + Kosky ch. 1 + Moaveni ch. 5). A professional code of ethics aims to ensure that the profession serves the legitimate purposes of all stakeholders: oneself, the employer, the profession and society. The code protects the members of the profession from the unwanted consequences of competition (the pressure to «cut corners» for the sake of savings), while leaving them free to enjoy its desirable consequences — invention and innovation. Armed with the code, an engineer can resist the pressure with the phrase: «As a professional, I cannot ethically place business interests above professional ethics».

The fundamental canons of the NSPE (National Society of Professional Engineers), the full version from Kosky 1.10.2 — engineers, in the fulfillment of their professional duties, shall:

1. Hold paramount the safety, health, and welfare of the public.
2. Perform services only in areas of their competence.
3. Issue public statements only in an objective and truthful manner.
4. Act for each employer or client as faithful agents or trustees.
5. Avoid deceptive acts.
6. Conduct themselves honorably, responsibly, ethically, and lawfully so as to enhance the honor, reputation, and usefulness of the profession.

Engineering Ethics: the Six Fundamental Canons of the NSPE

Week 1 · Kosky §1.10.2 · in the fulfillment of their professional duties, engineers shall:



Fig. 2. Engineering ethics (week 1): the six NSPE fundamental canons and the five cornerstones of personal ethics.

The Lab 1 adaptation of the ten canons (the handout «The 10 Canons of the Code», NSPE adapted) — the formulations are shorter, and these are the convenient ones to memorize: 1) Safety first — the safety, health and welfare of the public comes before everything; 2) Stay in your lane — work only within your area of competence; 3) Speak truthfully — public statements only objective and truthful; 4) No conflicts of interest — be a faithful agent, do not allow personal interest to conflict with duty; 5) No deception — avoid all deceptive acts; 6) Act honorably — behave responsibly, protecting the reputation of the profession; 7) No hidden payments — do not take payment from more than one interested party without full consent; 8) Don't misrepresent yourself — never falsify your qualifications, experience or results; 9) Give proper credit — acknowledge the real contribution of others; 10) Don't stay silent — report, or refuse to approve, violations of the norms, even under pressure.

The five cornerstones of personal ethical behavior (Five Cornerstones, Kosky 1.9.1): 1) do what you say you will do; 2) never disclose information given to you in confidence; 3) take responsibility for your actions (and mistakes); 4) never participate in a lie; 5) never accept gifts that would compromise your ability to act in the best interests of your organization. Kosky notes that the list is incomplete: it lacks the «golden rule» common to most ethical codes of the world («Do not do to others what you do not wish for yourself» — Confucius's version).

The top-10 questions to ask yourself when making a personal ethical decision (Kosky 1.9.2, counting from the 10th to the 1st): 10) Could the decision become a habit? If yes — don't do it. 9) Is it legal? If not — don't do it. 8) Is it safe? If not — don't do it. 7) Is it right? If not — don't do it. 6) Would it withstand public scrutiny? If not — don't do it. 5) If something terrible happens, will I be able to defend my actions? If not —

don't do it. 4) Is it fair, balanced and honest? If not — don't do it. 3) How will it make me feel about myself? If terrible — don't do it. 2) Does this choice lead to the greatest good for the greatest number of people? If not — don't do it. 1) And question No. 1: would I do this in front of my mother? If not — don't do it.

The Engineering Ethics Decision Matrix (Kosky 1.11) — the course's main tool. The rows of the matrix are the NSPE canons; the columns are possible courses of action (for example: go along with the decision; appeal to upper management; resign; write to your congressman; call a journalist). Each cell receives a brief answer to the question «does this particular course of action satisfy this particular canon?» (yes/no/maybe, with a note). Then the «yes» and «no» answers are summed across each column (ignoring the «maybes») — the criterion suggests the first action. The value of the matrix: it is «divide and conquer» — it forces you to consider all the dimensions instead of the first yes/no that comes to mind, and it prompts you to list all the relevant considerations. The exception: if one criterion overrides all the others — follow it.

The Engineer's Creed (NSPE 1954, from Moaveni ch. 5): «To give the utmost of performance; to participate in none but honest enterprise; to live and work according to the laws of man and the highest standards of professional conduct; to place service before profit, the honor and standing of the profession before personal advantage, and the public welfare above all other considerations». Moaveni also defines the terms: academic dishonesty (cheating, plagiarism, marking attendance for another person), plagiarism (presenting someone else's work as your own without citation), conflict of interest (a clash between personal interests and the duties of a position), contract (a voluntary agreement of two or more parties; legally binding), professional responsibility (the responsibility connected with possessing specialized knowledge and using it for the benefit of society).

2.3. Week-1 savollar (questions) — lecture and lab questions with model answers

Q: Lecture, interactive 1: «Why do you want to become an engineer?» (an answer in one or two words, not graded).

Answer: The model answer is built on three typical motives that the lecture itself suggests for reflection: (1) an inspiring person (a teacher, an engineer parent, a famous engineer); (2) a problem you wanted to solve (for example, power cuts made you want to understand power systems); (3) a favorite school subject (mathematics, physics). A strong answer ties the personal motive to the definition of engineering: «I want to be one of the people who turn scientific knowledge into working solutions».

Q: Lecture, interactive 2: «Which field of engineering interests you the most?» — why did you choose exactly this one?

Answer: Any answer is correct — this is an introduction to the audience. A model justification: name the field (for example, energy/electrical), connect it to specific products from the lecture's list (power, circuits, control systems) and to the NAE Grand Challenge it addresses (for example, «make solar energy economical»), and mention the professional society (IEEE).

Q: Lecture, discuss: «Name one ancient or historical structure that you consider a great engineering achievement. What problem did it solve?»

Answer: Model answer: the pyramids of Giza — the problem of the mass lifting and precise placement of millions of stone blocks without engines: solution = systems of ramps, levers and an organized workforce; the water aqueducts of Rome — the problem of supplying a growing city with water: solution = a continuous gradient of the channels over tens of kilometers, aqueduct bridges and settling tanks; city walls — the problem of defense (the oldest, military engineering). In the answer it is important to name the structure, state the problem and point to the engineering solution — following the lecture's «problem–solution» pattern.

Q: Lecture, interactive: «Describe in one sentence the engineering behind one ordinary moment of your day» (water, charging your phone, the bus).

Answer: Model: «Charging a phone is the result of engineering: the power plant (electrical engineering), the high-voltage lines and transformers (power engineering), the charging controller with overcharge protection (electronics), the connector and the fast-charging protocol — all of it designed, tested and standardized by engineers before I plugged in the cable».

Q: Lecture, interactive: «Which of the ten traits is your engineering super-skill?» — and why does a team need a mix of traits.

Answer: The trait is chosen individually; the justification of the choice is an example from experience (for example, vigilant: «in a physics project I was the first to notice an error in the data»). The answer to «why a mix»: the week-2 task is a team for the whole semester; the same trait in everyone creates blind spots: five «Objective» members argue endlessly and decide nothing, five «Creative» members generate ideas but no one checks them for safety; a team of different traits covers analysis, ideas, discipline and control.

Q: Lecture, discuss (repeated in Lab 1 and Kosky Example 1.3): «Your roommate spent the weekend partying and asks you to show him your homework before class. What do you do — and which canon/cornerstone applies?»

Answer: Kosky's canonical analysis via the five cornerstones: (1) «Do what you say» — by participating in an individual assignment you implicitly promised that the work is yours; showing the homework = breaking the promise. (2) «Do not disclose what is confidential» — homework is a confidential communication between student and instructor. (3) «Take responsibility» — by letting someone copy, you become an accomplice to the violation, which is itself an ethical violation. (4) «Do not participate in a lie» — copied work submitted as one's own is participation in a lie. (5) the one about gifts — not applicable (nothing was offered). Four cornerstones out of five support the answer d/e: refuse to show the homework; the «golden rule» adds empathy, so the best option is e — refuse, but offer to help them understand the material and do the work together. Under the TUT course code this is also a matter of academic honesty: first violation — 0 for the assignment.

Q: Lecture, interactive: «Describe in your own words the difference between a scientist and an engineer — in one sentence».

Answer: Model (your own formulation, not the textbook's): «A scientist asks why an airplane stays in the air, while an engineer builds an airplane that will not fall — and does it under the constraints of cost, weight and safety». Cross-check against Einstein: scientists investigate that which already is; engineers create that which has never been.

Q: Lab 1, Game 1 «The Ethics Courtroom», analysis of the example card: «A civil engineer is asked to sign the electrical wiring plan of a building because 'he is the only one free right now', although he has never studied electrical systems». What should be done, and which canon applies?

Answer: Verdict: decline and route the electrical work to a qualified electrical engineer. Canon No. 2 — Stay in your lane: perform services only in areas of their competence (the full NSPE wording). The answer format for any case in the game: verdict + canon number. Further arguments: an engineer's signature under an unsound plan also violates canon 1 (the safety of the public) and canon 8 (do not misrepresent your qualifications).

Q: Lab 1, Game 2 «Sort the Scenarios»: how do you sort the cards into Science / Engineering / Design / Overlap?

Answer: The lab's four bins: Science — investigates what already exists, the question «why does this happen?»; Engineering — creates what did not exist, «how do we make it work?»; Design — shapes the look, the look and feel, an open process without mandatory technical constraints; Overlap — design disciplined by physics, materials, cost and safety, that is, engineering design. The lab's tie-breaker: does the card take part in giving shape/appearance? No — Engineering. Yes, without technical constraints — Design. Yes, and with technical constraints — Overlap. Examples: «study why a bridge sagged» — Science; «design a bridge truss for the load» — Engineering/Overlap; «choose the color and shape of the body» — Design; «a body shape optimized for aerodynamics and safety» — Overlap.

Q: Lab 1, closing question: «Which idea from today's session will stay with you until next week?»

Answer: The model answer should name a specific idea and connect it to the coming project: for example, «engineering differs from repair in that it begins before anything has broken — with problem setting and analysis; next week we will start the systems analysis of our team system, and I will remember that the system boundary is an engineer's choice».

2.4. Savollar (questions) and topshiriqlar (assignments) from the textbooks for week 1 (with solutions)

Kosky, chapter 1 «What Do Engineers Do?» — the end-of-chapter exercises, written out in full; where the book gives an answer, it is shown in brackets.

1. Draw a conceptual sketch of your computer; use arrows and labels to indicate the keyboard, the screen, the power supply and the information storage devices.

2. Draw a conceptual sketch of a rugby ball and indicate all its components (the stitching, the cover, the bladder, the nipple, etc.) with arrows and numbers, as in Fig. 1.1.
3. Draw a conceptual sketch of a ballpoint pen, indicating all its components with arrows and labels.
4. Examine the exploded view of a desk: identify and label all its components.
5. Examine the exploded view of a box: identify and label all its components.
6. Redo Example 1.3 (the homework and the roommate) with the NSPE code instead of the cornerstones — solve it with the Engineering Ethics Decision Matrix. Solution: the rows of the matrix are the 6 NSPE canons; the columns are the actions (show it; show it with a warning; show it with a threat to report; refuse; refuse + help). By canon 2 (competence does not override the rules of the assignment) and canon 5 (do not participate in deception), the best columns are «refuse» and «refuse + help»; canon 1 is neutral here, canon 3 supports the honesty of the refusal.
7. Redo Example 1.4 (the bridge and the budget crisis) with the five cornerstones — solve it with the ethics matrix. The book's basic solution: summing the «yes/no» answers across the columns gives «appeal to upper management» (4 yes, 0 no) as the first action.
8. Final semester, dream job: you have accepted an offer from company A, but a dream offer has arrived from company B (a higher salary, a better company). What do you do about the accepted offer? Options: (a) simply not show up at A; (b) send A a letter withdrawing your acceptance; (c) ask B to call A and decline on your behalf; (d) turn down B and go to A as agreed. Model answer: (d) — faithful agent and honesty (canons 4 and 5); (b) is acceptable with a polite, immediate notification, but (a) and (c) are deception.
9. The company bought an expensive program for your summer job; the license allows a backup copy but use on only one computer at a time. Your professor wants to use it in your senior project. Options: (a) give the program to the professor; (b) copy it — «no one will find out»; (c) ask the company supervisor; (d) ask the professor to contact the company. Answer: (c) or (d) — obtain the permission of the rights holder; (a) and (b) violate the license (canons 5, 7, 10).
10. At a conference, a classmate is detained for drunkenness and jailed; he has no money for bail and does not want his parents to know; he asks to borrow the cash from the organization's till. Options: (a) lend it, since his parents are rich; (b) tell him to call his parents immediately; (c) give him the money against a written IOU; (d) tell him to call a lawyer — «it is not my problem». Model answer: (b) — do not spend other people's (the organization's) money without authority (canons 4, 7); the organization's money is not your property.
11. You are testing competitors' motorcycle helmets; your company has developed a cheap helmet that withstands multiple impacts but performs worse on the first impact than the competitor's; the vice president for sales demands the release and threatens dismissal. Options: (a) comply; (b) call the newspaper; (c) refuse to release an unsafe product and risk dismissal; (d) stall while you look for another job. Answer: (c) — canon 1 is paramount: the safety, health and welfare of the public is placed above everything else; before (b), the internal channels should be exhausted (canon 10 — do not stay silent about violations).

12. Paul, a manufacturing engineer at Bluestone Ltd., regularly meets with suppliers. One of them, Duncan, turns out to be a keen golfer and offers Paul a guest visit to the exclusive Cherry Orchard club, which Paul has always dreamed of visiting. Options: (a) accept — he has long wanted to; (b) decline, so as not to damage the business relationship; (c) ask to be nominated for membership of the club; (d) ask his supervisor whether the invitation may be accepted. Answer: (d) — ask about company policy: a gift from a supplier can compromise impartiality (canon 4 — faithful agent; cornerstone 5 about gifts).

13. Companies refuse to promote women to leadership positions in some countries, citing local customs and customer preferences. Discuss and defend your position. Model answer: the practice is discriminatory; gender must not be among the qualification requirements, and customer preferences do not justify discrimination (canons 3 and 6 — objectivity and the honor of the profession); it is also a matter of team diversity, which lecture 1 links to the quality of decisions.

14. Marvin Johnson, an environmental engineer, has prepared a report: the plant's discharge slightly exceeds the legal limit; there is almost no danger to people, at most harm to a small number of fish; fixing it would cost \$200,000. The supervisor asks him to adjust the data so that the plant looks compliant: «we cannot afford \$200,000, the competitors will pull ahead, tourism will be frightened off». How should Marvin respond? Model answer: refusal and an honest report — falsifying data violates canons 3 (objective and truthful statements), 5 (no deception) and 10 (do not stay silent about violations); the law is the law regardless of the cost of the solution; internal objections and the search for cheaper compliance are legitimate next steps.

15. Derek moved from a small company A (which did not require signing an assignment of software rights) to a large company B (which did). He was a key author of A's system; now at B he wants to adapt it for cross-references — «not selling it, just for myself; after all, I helped to create it, and I made changes». His friend Horatio asks: «Isn't that unethical? The system is A's property». Model answer: yes, it is unethical; the intellectual property belongs to A, and «small changes» do not change its origin; canons 4 (a faithful agent of the new employer), 7 (do not take what is not rightfully yours) and the confidentiality cornerstone apply; the right path is A's permission or writing your own system.

16. Jean, a professional engineer on unpaid leave, is a graduate student at a small private university and is enrolled for credit in a research course taught by Professor Dimanro; part of the research is an innovative geothermal technology. The university is expanding; the professor, a member of the building committee, is preparing a request for proposals and asks Jean to become a paid consultant to the committee that reviews the proposals. Jean's employer is not submitting a proposal and does not object. Is this a conflict of interest? (adapted from NSPE BER Case 91-5). Model answer: yes, this is a potential conflict of interest — a student enrolled for credit in the course is at the same time evaluating proposals that affect the technology of her own research; canon 4; it is permissible only with full disclosure to the interested parties and the consent of all of them (following the logic of the BER decision).

Moaveni, chapter 1 «Introduction to the Engineering Profession» — the end-of-chapter problems, written out in full:

1. 1.1. Observe your surroundings: which engineering achievements available to you today could you not do without?

2. 1.2. Find a suitable professional organization for one of the disciplines (bioengineering, ceramic, chemical, civil, computer, electrical, electronic, environmental, industrial, manufacturing, materials, mechanical engineering) on the internet and prepare a two-page report on the organization's goals and mission.
3. 1.3. To raise society's awareness of engineering, prepare and deliver a 15-minute web presentation at a mall in your city (with advance planning and permission).
4. 1.4. If your course has a semester project — present the final work at a mall on the appointed day; if there is a competitive component, hold the design competition there.
5. 1.5. Prepare a 15-minute oral presentation on engineering and its disciplines and present it to the high school students of your hometown school.
6. 1.6. Class project: a website about engineering and its branches; choose a group leader and distribute the tasks; along the way, record the pleasures and problems of teamwork; write a brief report with recommendations.
7. 1.7. Team project: a web presentation on the history and the future of engineering; collect pictures, short videos and charts; give links to the main engineering societies and R&D; centers.
8. 1.8. Class project: ask your parents to think back to their graduation year and compile a list of products and services that are available now but were unavailable then (mobile phones, ATMs, personal computers, airbags, price scanners, etc.); ask whether they imagined these things back then and how these products have improved (or worsened) life.
9. 1.9. Class project: compile a list of products and services that do not exist now but that, in your opinion, will become available in the next 50 years; combine the lists and present them to the class.
10. 1.10. Perform a web search on the number of engineers by field and their average salaries over recent years; present the findings to the instructor.
11. 1.11. If you plan a career in chemical engineering: find out what polymers, plastics, thermoplastics and thermosetting plastics mean; give ten examples of everyday plastic products; a brief report.
12. 1.12. If you plan a career in electrical engineering: find out how electricity is generated and distributed; a brief report.
13. 1.13. Electric motors surround us: name at least ten household products with electric motors.
14. 1.14. Name at least 20 different materials used in household products.
15. 1.15. If you plan a career in civil engineering: find out what dead load, live load, impact load, wind load and snow load mean in structural design; a short note to the instructor.
16. 1.16. Group project: society depends heavily on engineers — the mistakes of engineers cost money and lives. «An incompetent and unethical surgeon can cause the death of at most one person at a time, whereas an incompetent and unethical engineer can cause the death of hundreds at once». Visit the NSPE website, study engineering ethics, discuss why it is so important and why engineers are obliged to practice to the highest standards of integrity; give examples of codes; a report to the instructor.

Moaveni, ch. 1, the built-in activity «Impromptu Design I» (a mini-competition): design a device from the issued materials (2 paper plates, 1 Dixie cup, 2 balloons, 3 rubber bands, 1 straw, 12 self-stick labels) so that the device falls from a height of 10 feet and lands inside a marked 4×4 ft zone; use all the items; one practice attempt is allowed; the design with the slowest fall time wins; 30 minutes for the preparation. This is the first encounter with a design competition (rules, constraints, the winning criterion).

Moaveni, chapter 2 «Preparing for an Engineering Career» — the five keys to a student's success (they echo Kosky 1.7; there is no need to compare them word for word, the meaning is the same: read the book before class; do all the homework — «homework is your engineering practice»; attend every class; treat studying as a job — 2–3 hours of independent work for every hour of lectures; the responsibility for learning is yours). Chapter assignments: 2.1 — draw up a semester schedule and two alternatives, discuss their pros and cons, choose the best one, agree on it with your adviser, keep a daily log of deviations, and every two weeks submit a summary and improvements; 2.2 — meet with the adviser and prepare a graduation plan. Chapter summary: a smooth transition from school to university, time management from the first day, attendance, timely help, good lecture notes, a place to study, study groups; the importance of joining an engineering organization and participating actively.

2.5. Atamalar (terms) — Week-1 glossary

Term (EN)	Atama (UZ)	One-phrase definition
Engineering	Muhandislik	The practice of using natural science, mathematics and the design process to solve technical problems and improve systems
Engineer	Muhandis	A creative, inventive person who finds solutions to practical problems (from Latin ingenium/ingeniare)
Civil engineering	Qurilish muhandisligi	One of the oldest disciplines; from civilian engineers — the first non-military engineers
Military engineering	Harbiy muhandislik	The oldest form of engineering: fortifications, siege works, roads for armies
Engineering design	Injenerlik dizayni	Design disciplined by analysis: physics, materials, cost, manufacturability, safety
Professional society	Kasbiy jamiyat	A discipline's organization: standards, continuing education, a code of ethics (ASME, IEEE, ASCE, AIChE, AIAA, ANS)
Code of ethics	Axloq kodeksi	A body of rules ensuring that the profession serves the interests of society, the employer, the profession and oneself
Canon	Kanon	A rule of acceptable conduct for a particular group (historically, church laws)
Personal ethics	Shaxsiy qadriyatlar	Standards of behavior by which people judge the morality of personal and group situations

Term (EN)	Atama (UZ)	One-phrase definition
Engineering Ethics Decision Matrix	Axloqiy qarorlar matritsasi	A «canons × courses of action» table for working through ethical dilemmas
Faithful agents or trustees	Sadoqatli vakil/aminomonish	Acting in the interest of the employer/client without a conflict of personal interest
Grand Challenges for Engineering	NAE Grand Challenges	The 14 social missions of 21st-century engineering (clean water, infrastructure, medicine, solar energy, cybersecurity, better medicines, etc.)
Technologist	Texnolog	A graduate of an engineering technology program; implementation and extension of existing technologies
Technician	Texnik	A graduate of a 2-year program; quality control, testing, data collection
Engineering sketch	Muhandislik eskizi	A free, quick drawing of an idea, roughly proportional to the object

Table 2.1. Week-1 terms (English — Uzbek)

3. Week 2 — Systems Thinking I: systems, boundaries, stakeholders

3.1. Key points (“all qoidalar” of week 2)

A recap of week 1 in one breath (the slides themselves open this way): engineering is a creative, inventive response to practical problems, not repair work; it has a history — from ancient walls and aqueducts to steam, electricity, flight, and software; a social role — engineers sit at the interface between science and society; an economic role — one of the world's largest and best-paid professions; and a common culture — professional societies, shared traits, and a code of ethics.

Definition of a system (the key definition of the week): a system is a set of interacting elements and processes that transform inputs into outputs and exists within an operating context. The four parts of the definition: elements (the parts and the people that make up the system), processes (how the parts interact and transform things over time), inputs and outputs (what flows in and what flows out), and the operating context (the conditions under which the system exists).

A system is NOT a pile of parts (NOT THE SAME AS). A random heap of parts is not a system; a system is parts with a purpose, connected so that they act together rather than simply lie together. Four “not a system / system” pairs from the lecture: a pile of bricks in the yard — a brick wall (bricks + mortar + a load path, holding the weight together); a student, a timetable, and a bus stop taken separately — a university bus route (vehicles, schedule, and passengers interact to move people); a stove, a refrigerator, and a sink standing in a room — a cafeteria kitchen (staff, equipment, and a workflow that turns ingredients into meals); a phone, a

bank, and a shop with no link between them — a mobile payment app (accounts, verification, and transfer tied into one flow).

The system boundary. The boundary is a drawn line, often imaginary, separating “inside the system” from “outside”. Anything may cross it: materials, people, money, information, energy. The key idea of the week: **the boundary is an engineer's choice, not a fact about the world**. You can draw it tightly around one machine or broadly around an entire building — each choice changes what has to be analyzed. Lab 2 repeats the same rule: “the boundary is drawn, not discovered”; two engineers may draw the boundary of the same system in different places — both choices are valid, provided the choice is justified and applied consistently.

The operating context. Just outside the boundary sits the operating context — conditions that do not cross the boundary as inputs yet still constrain the system's behavior. The four categories from the lecture: codes and standards (sanitary, safety, building); time and schedule (opening hours, the academic calendar, deadlines); existing infrastructure (what is already built and with which the system must coexist); budget and resources (what funding, staffing, and materials are actually available). The cafeteria example worked through: inputs — food deliveries, staff arriving, students coming in; outputs — meals served, food waste, payments; context — the sanitary code, opening hours, the kitchen budget, the building layout.

Stakeholders. A stakeholder is anyone who uses the system, runs it, or is affected by it — whether or not they sit inside the boundary. The three columns of the stakeholder map: Uses It — uses it (students eating in the cafeteria; bus passengers); Runs It — runs it (kitchen staff, dispatchers, maintenance crews); Affected By It — affected by it (neighbors living with the noise, the trucks, the congestion — often entirely outside the boundary). The stakeholder map is a simple table; “a first draft is enough to get started”. Importantly, stakeholders do not always want the same thing. Good systems thinking notices the tension instead of hiding it. The lecture example: the university wants the bus route to be faster; the maintenance crew — safer; the passengers — cheaper. Question: who are the stakeholders here, and where is the tension?

System Anatomy: boundary, inputs-outputs, context, stakeholders

Week 2 · the course example — a university cafeteria

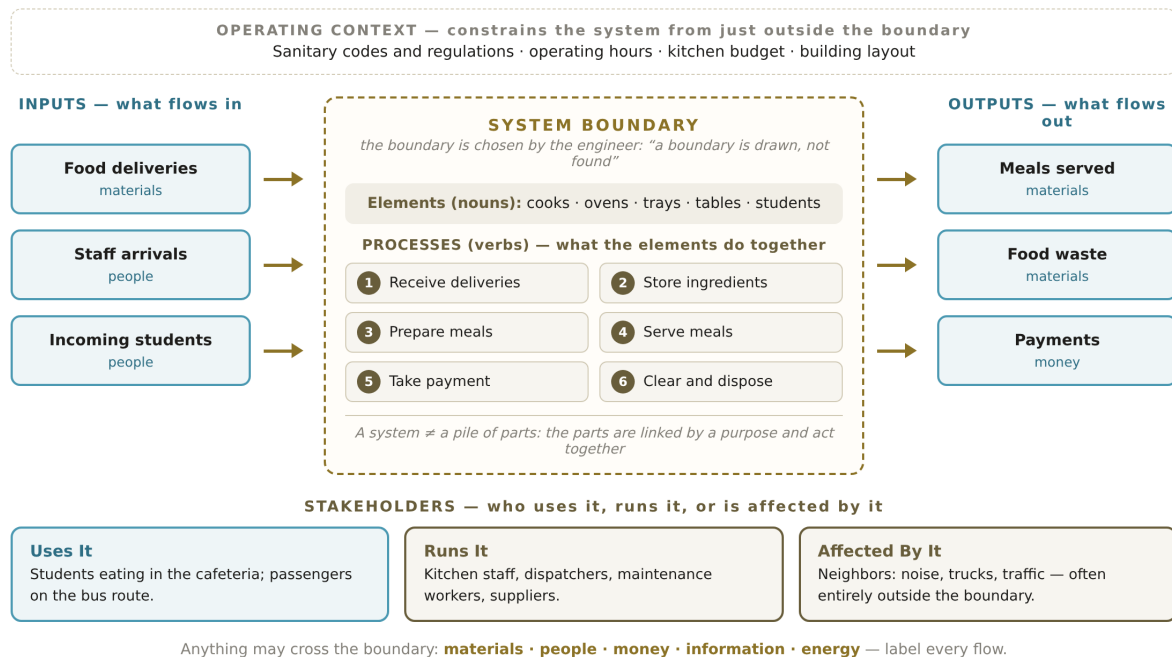


Fig. 3. Anatomy of a system (week 2): boundary, inputs and outputs, operating context and stakeholders, using the university cafeteria example.

A bridge to the project — Milestone 1 (the assignment announced in lecture 2): 1) finalize the system — choose one system the team will study for the whole semester (it can be the one chosen at the end of lecture 1); 2) sketch the boundary — draw the boundary line and list at least 3 inputs and 3 outputs across it; 3) map the stakeholders — sort them into Uses It / Runs It / Affected By It. The deadline is before lecture 3; this is a draft of Milestone 1, and at the next session (Systems Thinking II) it turns into a process flow diagram. Course idea prompts: a dorm laundry, a university bus route, the campus cafeteria, a mobile payment app.

3.2. Week-2 savollar (questions) — lecture and lab questions with model answers

Q: Lecture: “A system or just a collection?” — classify: (a) a pile of bricks; (b) a bus route; (c) the appliances in a kitchen; (d) a payment app.

Answer: (a) not a system — the parts lie around with no connection and no purpose; (b) a system — vehicles, schedule, and passengers interact with the purpose of moving people; (c) the appliances on their own are not a system, but the cafeteria kitchen (staff + equipment + workflow) is; (d) a phone, a bank, and a shop separately are not a system, but the payment app ties accounts, verification, and transfer into one flow — a system. The criterion: are there interacting elements and processes transforming inputs into outputs for a purpose.

Q: Lecture (homework question): “Take one of your three homework systems. Where would you redraw its boundary — tighter or wider?”

Answer: Model answer (example: the dorm laundry): tighter — only the machines and the drop-off counter: then the electricity suppliers and repair services become external counterparties, and the analysis focuses on the queue at the machines; wider — the whole linen chain: student rooms, hampers, drying, scheduling — then resident behavior and the daily schedule enter the analysis. There is no single right answer: you must justify the choice explicitly and keep it consistent — that is exactly “the boundary is an engineer's choice”.

Q: Lecture, the interactive “Inside or outside the boundary?” (cafeteria): (1) the supplier's truck; (2) the meal-plan billing office.

Answer: The official answer was given in lecture 3: (1) the supplier's truck is usually outside: deliveries cross the boundary as an input at step 1; the truck is a stakeholder supplying the system, but none of our steps happen to the truck itself. (2) the meal-plan billing office depends on where you drew the line: outside, if the diagram ends at the “Take payment” step; inside, if it includes “Verify meal plan” and “Update balance”. “Any answer can be defended, as long as you declared the boundary and keep it consistent”.

Q: Lecture, Discuss: “The university wants it faster, the maintenance crew — safer, the passengers — cheaper. Who are the stakeholders of the bus route, and where is the tension?”

Answer: Stakeholders: passengers (Uses It), drivers/dispatchers and the maintenance crew (Runs It), the university administration, neighbors along the route (Affected By It). Tension: speed against safety (faster = tighter schedules = fatigued drivers and risks), speed against price (faster = more buses = a higher fare or budget), safety against price (maintenance and inspections cost money). Model takeaway: a conflict of interests is a property of the system, not a flaw in someone's map; it must be shown in the stakeholder map, not hidden.

Q: Lab 2, Game 1 “Reverse Engineer an Everyday System”: the rules and the reference analysis of the “traffic light intersection” card.

Answer: Game rules: the team draws a card with an everyday system (a vending machine, a traffic light, a coffee shop); breaks the system down into its key elements and the processes linking them; draws a boundary line and lists at least 3 inputs and 3 outputs; a speaker explains the system in 60 seconds. The reference card “Traffic Light Intersection”: inputs — vehicle and pedestrian sensor data, electrical power, the signal plan/schedule; outputs — red/yellow/green signals, traffic-flow data, the “walk” signal for pedestrians; operating context — the city traffic code, the rush-hour schedule, the existing road geometry, the municipal maintenance budget.

Q: Lab 2, Game 2 “Stakeholder Detective”: the reference analysis of the “university makerspace (3D-printing room)” card and its tension.

Answer: Uses It — students printing course projects; club teams printing parts. Runs It — lab technicians, student assistants, the procurement office. Affected By It — neighbors in the building (noise, fumes), the fire safety officer. Tension: students want round-the-clock access; the fire officer wants supervision and a night shutdown; procurement wants a spending cap on filament. “Who is right depends on whose goals you weigh more heavily — that is exactly the judgment the group work is meant to exercise”.

Q: Lab 2, graded group work “Applied Systems Challenge”: six steps on one card — name the steps.

Answer: 1) Name the system — a precise name following the principles of lecture 2. 2) Draw the boundary — what is inside/outside and the justification. 3) List the inputs and outputs — at least 3+3 across the boundary. 4) Stakeholder analysis — Uses It / Runs It / Affected By It. 5) An agreed solution — one resolution of the card's conflict, and a note on whom it pushes forward. 6) Operating context — codes, schedule, infrastructure, budget. Grading is live: the instructor walks around with a checklist and scores all six steps before time runs out.

Q: Lab 2, model answer for the “Campus EV Charging” card (an electric-vehicle charging station).

Answer: Situation: 8 charging ports are planned; students want more; facilities fear peak demand on the grid; the offices want to keep their parking. 1) System: Campus EV Charging Station Network. 2) Boundary: the 8 charging boxes + the parking zone. 3) Inputs: electrical power, arriving EVs, charging-station equipment; outputs: charged cars, paid-for energy, bay occupancy data. 4) Stakeholders: Uses It — students and staff driving EVs; Runs It — facilities and maintenance; Affected — neighboring offices, people parking conventional cars. 5) Solution: 8 chargers in the two least-loaded lots; a 2-hour charging limit at peak (9:00–17:00) to manage grid load; 2 EV-only bays excluded from general parking, protecting office access. 6) Context: campus grid capacity, the university's parking policy, the budget for installing 8 chargers, the peak-pairs schedule (9:00–17:00). Note: this is a model answer — a different well-argued solution scores just as highly.

3.3. Week-2 savollar (questions) and qoidalar (rules) from the textbooks

Kosky, chapter 2 “Elements of Engineering Analysis” — the basic vocabulary of analysis on which systems thinking rests: an engineering calculation differs from a purely mathematical one in that a single number is not enough. The result of an engineering analysis must be right in four ways (the chapter summary): 1) involve the relevant variables; 2) have the correct numerical value; 3) be expressed in the correct units; 4) carry the correct number of significant figures. A variable (2.3) is a quantity that can be measured: “how hot?” — temperature, “how heavy?” — weight, “what voltage?” — electrical potential; variables are almost always defined through measurements made with familiar instruments — thermometers, rulers, clocks; speed, for example, is a ruler measurement (distance) divided by a clock measurement (time).

Units of measurement (2.4): a calculated quantity always has two parts — a numerical value and units; the answer must be right in both categories. Units quantify the concept of dimensions (dimensions): dimensions

are the fundamental quantities we perceive (mass, length, time), while units provide the numerical scale for measuring them; units are established arbitrarily and fixed by civil law or custom — how length is measured (feet or meters) has nothing to do with the laws of physics. The lecture-book example: the “100” on a Canadian highway speeding ticket — 100 mph or 100 km/h? 100 km/h is only 62 mph; units matter.

The SI system (2.4.1): the metric standard, based on MKS (meter, kilogram, second). The fundamental SI units in Kosky: the meter (m) — length, the second (s) — time, the kilogram (kg) — mass, the kelvin (K) — temperature, the mole (mol) — the amount of particles, the ampere (A) — electric current. Moaveni (ch. 6) adds the full list of seven base dimensions: length, mass, time, temperature, electric current, amount of substance, luminous intensity (candela), plus the formal BIPM definitions: the meter — the path traveled by light in vacuum in $1/299,792,458$ of a second; the second — 9,192,631,770 periods of the radiation of the cesium-133 transition; the kilogram — the mass of the international prototype; the kelvin — $1/273.16$ of the thermodynamic temperature of the triple point of water; the mole — the amount of substance containing as many elementary entities as there are atoms in 0.012 kg of carbon-12; the ampere — the current that, in two parallel conductors 1 m apart, produces a force of 2×10^{-7} newton per meter; the candela — the luminous intensity of a source with a frequency of 540×10^{12} Hz and a radiant intensity of $1/683$ watt per steradian.

Qoidalar for writing units (Kosky 2.4.2): 1) all unit names are written in lowercase even when formed from names (newton, joule) — except at the start of a sentence; 2) the abbreviation of a unit formed from a person's name is capitalized (N, J); 3) a second letter appears in an abbreviation only to avoid confusion with other units (Wb — weber, to distinguish it from W — watt; k in kW); 4) abbreviations never take a plural: kilograms — kg, not kgs; newtons — N, not Ns; seconds — s, not sec.; 5) an abbreviation takes no period at the end (except at the end of a sentence); 6) units not formed from names are written lowercase in both the name and the abbreviation: meter (m), kilogram (kg), second (s). Kosky's fail-safe units method: in calculations the numerical value always carries square brackets with the units: 3 [in]; conversion factors are written as 12 [in/ft]; converting 10 feet: $(10 \text{ [ft]}) \times (12 \text{ [in/ft]}) = 120 \text{ inches}$; the final answer carries no brackets.

Derived SI units (tables 2.3 in Kosky / 6.3 in Moaveni): frequency — hertz (Hz) = $1/\text{s}$; force — newton (N) = $\text{kg} \cdot \text{m}/\text{s}^2$; energy — joule (J) = $\text{N} \cdot \text{m} = \text{kg} \cdot \text{m}^2/\text{s}^2$; power — watt (W) = J/s ; electric charge — coulomb (C) = $\text{A} \cdot \text{s}$; electric potential — volt (V) = W/A ; resistance — ohm (Ω) = V/A ; capacitance — farad (F) = C/V ; pressure/stress — pascal (Pa) = N/m^2 ; rotational frequency, torque ($\text{N} \cdot \text{m}$), momentum ($\text{N} \cdot \text{s}$), quantity of motion ($\text{kg} \cdot \text{m}/\text{s}$), density (kg/m^3), magnetic flux (Wb), inductance (H). SI prefixes (table 2.4 in Kosky / 6.2 in Moaveni): exa (E, 10^{18}), peta (P, 10^{15}), tera (T, 10^{12}), giga (G, 10^9), mega (M, 10^6), kilo (k, 10^3), hecto (h, 10^2), deka (da, 10^1); submultiples: deci (d, 10^{-1}), centi (c, 10^{-2}), milli (m, 10^{-3}), micro (μ , 10^{-6}), nano (n, 10^{-9}), pico (p, 10^{-12}), femto (f, 10^{-15}), atto (a, 10^{-18}).

Other systems of units (Moaveni 6.2): British Gravitational (BG) — the foot (0.3048 m), the second, and the pound-force lb (4.448 N) as the base unit, with the slug as the derived unit of mass: $1 \text{ lb} = (1 \text{ slug}) \cdot (1 \text{ ft}/\text{s}^2)$; temperature — $^{\circ}\text{F}$ or $^{\circ}\text{R}$. U.S. Customary — the foot, the second, and the pound mass lbm (0.453592 kg); the lbf is defined as the weight of an object of mass 1 lbm at sea level at 45° latitude, where $g = 32.2 \text{ ft}/\text{s}^2$; $1 \text{ lbf} = 4.448 \text{ N}$; the relation $1 \text{ slug} = 32.2 \text{ lbm}$. Temperature formulas: $T(^{\circ}\text{R}) = 9/5 \cdot T(\text{K})$; $T(^{\circ}\text{F}) = 9/5 \cdot T(^{\circ}\text{C}) + 32$; $\text{K} = ^{\circ}\text{C} + 273.15$ (Moaveni writes 273.16 in the definition of the kelvin via the triple point); $^{\circ}\text{R} = ^{\circ}\text{F} + 459.67$. Key conversion factors: $1 \text{ ft} = 0.3048 \text{ m}$; $1 \text{ m} = 3.2808 \text{ ft}$; $1 \text{ lbm} = 0.4536 \text{ kg}$; $1 \text{ kg} = 2.2046 \text{ lbm}$; 1

lbf = 4.448 N; 1 J = 0.7375 ft·lbf; 1 ft·lbf = 1.3558 J; 1 Btu = 778.17 ft·lbf; 1 hp = 550 ft·lbf/s = 0.7457 kW; 1 W = 0.7375 ft·lbf/s.

The cautionary case of the week (Kosky exercise 42 and Moaveni 6.3): the Mars Climate Orbiter, launched on December 11, 1998, was lost on September 23, 1999 while entering orbit at Mars. The commission concluded: NASA engineers failed to convert the rocket thrust from pounds-force to newtons (the units of the navigation software), so the spacecraft entered the Martian atmosphere at an altitude of about 57 km instead of the planned 140–150 km and was destroyed. The lesson: inconsistent units kill missions; check unit consistency across the entire system.

Kosky, chapter 2, exercises on the week's topics (Exercises, a selection by the themes of weeks 2–4; the full block 1–44 also includes drawings, unit conversions, and spreadsheets):

1. 4. If a US gallon = 0.134 ft^3 and the volume of a human mouth = 0.900 in^3 , how many swallows of water are needed to fill a 5.00 (US) gallon canister? (Book answer: 1.29×10^3 swallows.)
2. 5. Determine how to perform the conversions — by definition, via conversion factors, by geometry, or by a scientific law: (a) how many square miles in a square kilometer; (b) how many microfarads in a farad; (c) the weight, in newtons, on Earth of an object of mass 10 kg; (d) how many square miles on the surface of the Earth.
3. 6. The acre was originally the area a team of oxen would plow in a day: 0.040 ha/day ; a hectare is exactly 10^4 m^2 ; there are 1609 meters in a mile. How many acres in a square mile?
4. 7. There are 39 inches in a meter. What is the area, in SI, of the skin of a spherical orange of diameter 4.0 inches? (Answer: $3.3 \times 10^{-2} \text{ m}^2$.)
5. 8. What is the volume, in the Engineering English system, of a spherical apple of diameter exactly 10 cm (note the decimal point)?
6. 9. If the ranch of Example 2.3 were circular rather than square, for the same money (\$320,000; fencing at $\$10 \times 10^3/\text{mile}$; land at $\$10 \times 10^4/\text{square mile}$) — what is the diameter of the ranch?
7. 10. A Fermi problem: estimate the number of piano tuners in Chicago (population 5 million). (Book answer corridor: plausibly 50–250 tuners.)
8. 13. If the mass in Example 2.5 is increased from 10.0 kg to 20.0 kg and the stretch doubles, how far will a 4.00 m wire stretch under that mass?
9. 17–19. Gasoline and the weight of a car: $G = W^{2/3}$ (G — the weight of gasoline in lbf, W — the weight of the car in lbf); gasoline weighs 8.0 lbf/gallon; fuel economy mpg = $84500/(\text{weight in lbf}) - 3.0 \text{ mpg}$. 17) the fuel economy of a $3.00 \times 10^3 \text{ lbf}$ car? 18) the heaviest car achieving a range of 600 miles? (Answer: $3.7 \times 10^3 \text{ lbf}$.) 19) a plot of the range as the exponent b varies from 0.50 to 0.75.
10. 24–26. The cost of operating a car: annual fixed costs $\approx 20\%$ of the price; cost/mile = $0.20/\text{year} \times (\text{price})/(\text{miles per year}) + (\text{gasoline price/gallon}) \times (\text{gallons/mile})$; 2.0×10^4 miles per year, gasoline at \$5.00/gallon. 24) the cost/mile of a \$15,000 car with 30.0 mpg? (Answer: \$0.32/mile.) 25) what mpg would a car twice as expensive need for the same cost per mile?
11. 31–33. Hubbert's Peak (oil): the world had 3000×10^9 barrels; exploration since 1850; every 25 years 10% of what remains is found; consumption 1000×10^5 barrels in 1850, growing fivefold every 25 years.

When do the reserves turn negative? (Answer: the year 2025.)

12. 35. Compute with the correct significant figures: (a) $100/(2.0 \times 10^2)$; (b) $1.0 \times 10^2/(2.0 \times 10^2)$.

(Answer: 0.5 and 0.50.)

13. 36. With the correct significant figures: 10/6; 10.0/6; 10/6.0; 10./6.0; 10.0/6.00.

14. 37. What is 2.68×10^8 minus 2.33×10^3 with the correct significant figures? (Answer: 2.68×10^8 .)

15. 39. Round to three significant figures: 1.53; 15.345; 16.67; 102.04; -124.7; 0.00123456.

16. 40. A door and its frame: to how many significant figures should the length and width of a 30.0" × 81.0" door be specified, so that it keeps out the draft and still opens easily?

17. 42. Mars Climate Orbiter (full text above): as NASA's chief engineer, how do you respond to the national outcry over the "silly" mistake? Options: (a) take all the blame and resign; (b) find the culprit and fire/demote them; (c) audit the software for compliance with the specifications across all data exchanged between working groups; (d) check the consistent use of units across the whole design and operations. Model answer: (c) and (d) — a systemic fix of the process; the NSPE ethics matrix is attached.

18. 43. A classmate sent you their completed spreadsheet homework together with the answer to a question. What do you do? Options: (a) delete it without looking; (b) peek to see "whether they did it right"; (c) paste it into your own homework, changing the formatting; (d) send it on to all your friends. Model answer: (a) — zero tolerance for academic dishonesty; looking already pulls you toward using it.

19. 44. Stephanie knew that Adam, the environmental manager, would be displeased with her report on a chemical spill: the data clearly showed the spill was large enough that state regulations required reporting it. Adam flared up: "A couple of gallons over the limit is not worth filling out the damn forms; go redo the numbers until they come out right." What should Stephanie do? Options: (a) say that she will not knowingly violate state law and threaten to quit; (b) comply — the boss is the one accountable; (c) an anonymous report to the state; (d) go over Adam's head — to his supervisor. Model answer: (a) plus escalation (d) if refused; falsification is forbidden by canons 3, 5, 10.

Moaveni, chapter 6 "Fundamental Dimensions and Units" — end-of-chapter assignments related to the themes of weeks 2–4 (unit conversion, significant figures, systems and components):

1. 6.1–6.2. Convert a table of quantities: SI to U.S. Customary — 120 km/h to mph and ft/s; 1000 W to Btu/h and hp; 100 m^3 to ft^3 ; 80 kg to lbm; 1000 kg/m^3 to lbm/ft^3 ; 900 N to lbf; 100 kPa to lbf/in^2 ; 9.81 m/s^2 to ft/s^2 . And back: 65 mph to km/h and m/s; 60,000 Btu/h to W; 120 lbm/ft^3 to kg/m^3 ; 30 lbf/in^2 to kPa; 200 lbm to kg; 200 lbf to N. Show all steps.

2. 6.3. The angle of twist of a shaft: $\varphi = TL/(JG)$, where T — torque ($\text{N}\cdot\text{m}$), L — length (m), G — shear modulus (N/m^2). What must the unit of J be for the equation to be homogeneous in units? Answer: m^4 (the polar moment of inertia).

3. 6.4. Which equations are dimensionally homogeneous? Prove it: (a) $F = ma$; (b) $F = mV^2/R$; (c) $F(t_2 - t_1) = m(V_2 - V_1)$; (d) $F = mV$.

4. 6.5. Determine the number of significant figures: 286.5; 2.2×10^2 ; 2200; 0.0286.

5. 6.6. Write the results with the correct number of significant figures: $1.2856 + 10.1$; $155 - 0.521$; $155 - 0.52$; 1558×12 ; 3.585×1.2 .
6. 6.7–6.8. For the systems (a jacket, trousers, a skirt, shoes, a bicycle, roller skates; a refrigerator, a computer, the human body, a building, a water heater, a toaster, an airplane) identify the main components and the role of each — practice in decomposing systems into elements.
7. 6.9. What do these laws describe: Fourier's, Darcy's, Newton's viscosity, Newton's cooling, Coulomb's, Ohm's, the ideal gas, Hooke's, the first law of thermodynamics, Fick's, Faraday's?
8. 6.13. Dimensional homogeneity: (a) $F(t_2 - t_1) = mV_2 - mV_1$; (b) $F(V_2 - V_1) = \frac{1}{2}mx_2^2 - \frac{1}{2}mx_1^2$; (c) $F = \frac{1}{2}mV_2^2 - \frac{1}{2}mV_1^2$; (d) $F(x_2 - x_1) = \frac{1}{2}mV_2^2 - \frac{1}{2}mV_1^2$.
9. 6.14. A car of mass 1500 kg: express the mass and the weight in BG and U.S. Customary, showing the steps.
10. 6.15. Express the kinetic energy (mass)(speed)² of a 1200 kg car at 100 km/h in SI, BG, and U.S. Customary.
11. 6.16. A workshop floor of 30 ft × 50 ft: express the area in ft², m², in², cm².
12. 6.17. A trunk of 18 ft³: express it in in³, m³, cm³.
13. 6.18. An Acura RL: 300 horsepower, a 3.5-liter engine — express in kW and in³.
14. 6.19. The density of air at room conditions is 1.2 kg/m³: express it in BG and U.S. Customary.
15. 6.20. In Phoenix it is 68°F inside and 110°F outside: the temperature difference in (a) °F, (b) °R, (c) °C, (d) K? Is one degree of difference in Celsius equal to one degree in Kelvin, and in Fahrenheit to one in Rankine? Why?
16. 6.21. A cantilever beam: the deflection $y = wx^2(x^2 - 4Lx + 6L^2)/(24EI)$; what must the unit of w (the distributed load) be for the equation to be homogeneous? Answer: N/m.

3.4. Week-2 atamalar (terms) — glossary

Term (EN)	Atama (UZ)	One-phrase definition
System	Tizim	Interacting elements and processes that transform inputs into outputs, within an operating context
Element	Element	A part or a person making up the system
Process	Jarayon	How the elements interact and transform things over time
Input / Output	Kirish / Chiqish	Flows across the boundary: materials, people, money, information, energy
System boundary	Tizim chegarasi	A drawn (often imaginary) inside/outside line; an engineer's choice, not a fact about the world
Operating context	Ish konteksti	Conditions just outside the boundary that constrain behavior: codes, schedule, infrastructure, budget

Term (EN)	Atama (UZ)	One-phrase definition
Stakeholder	Manfaatchi (stakeholder)	Anyone who uses the system, runs it, or is affected by it — even from outside the boundary
Stakeholder map	Manfaatchilar xaritasi	Sorting stakeholders into Uses It / Runs It / Affected By It; the first draft is enough
Tension	Qarama-qarshilik (tension)	A conflict of stakeholder wants (faster/safer/cheaper) that must be surfaced, not hidden
Variable	O'zgaruvchi	A measurable quantity: a number plus units (temperature, weight, voltage)
Dimension	O'lcham (dimension)	A fundamental perceived quantity (mass, length, time) that units quantify
SI	SI tizimi	The international system of units based on MKS: m, s, kg, K, mol, A (+ cd)
British Gravitational	British Gravitational	The system with the pound-force as the base unit and the derived slug: $1 \text{ lb} = 1 \text{ slug} \cdot \text{ft/s}^2$
U.S. Customary	U.S. Customary	The system with lbm and lbf; the lbf is the weight of 1 lbm at $g = 32.2 \text{ ft/s}^2$
Milestone 1	Milestone 1	The project's first milestone document: chosen system + boundary + stakeholder map

Table 3.1. Week-2 terms (English — Uzbek)

4. Week 3 — Systems Thinking II: process flow diagrams (PFD)

4.1. Key points (“all qoidalar” of week 3)

A recap of week 2 (in one breath): a system is interacting elements and processes turning inputs into outputs; the boundary is a line the engineer chooses; the operating context constrains the system from just outside the boundary; stakeholders are those who use it, run it, or are affected by it. Today — what happens inside the boundary, and who sees it how.

What a PFD is: a process flow diagram shows the steps of a process in order and what flows between them. Three things it gives you: the sequence (what happens first, next, and last); the flow (what moves between the steps: material, people, information, money, energy); the problems (gaps, waits, loops, and hand-offs become visible). Kosky et al. apply the same idea to the design process itself (Fig. 21.1): the design process is a process too.

Elements are nouns, processes are verbs. The cafeteria's elements: cooks, ovens, trays, tables, students. The processes: receive, store, prepare, serve, take payment, clear. Lecture 2 listed the parts; the PFD shows what

they do together.

One process — three levels of detail (rule: choose the level to fit the question, and keep one level per diagram). Level 0 (black box): inputs come in, outputs go out, the inside is hidden — the cafeteria: food, staff, and students come in; meals, waste, and payments go out. Level 1 (main steps): a handful of main steps inside the boundary — Receive deliveries → Store ingredients → Prepare meals → Serve meals → Take payment → Clear and dispose. Level 2 (zoom in): one level-1 step expanded into its own diagram — “Prepare meals”: wash → chop → cook → hold hot.

The five symbols of the PFD language (course convention): Start/End — where the process begins and ends; Step — a single action, named “verb + noun”; Decision — a question with labeled Yes/No exits; Flow — an arrow for the order of the steps and for what moves; Wait/queue — where things sit and wait (course convention — a dashed box). “Five symbols are enough for the course; the sequence matters more than the particular choice of symbols”.

Five rules for a readable diagram (memorize them): 1) Start and end explicitly: one start, and every path reaches an end. 2) Name the steps “verb + noun”: “Receive deliveries”, not “Delivery”. 3) Label the arrows: say what moves — food, students, information, money. 4) Give every decision both exits: every Yes and every No must lead somewhere. 5) Keep one level of detail: arrows crossing the boundary are your inputs and outputs.

Every arrow carries something (five categories — the same as for the week-2 inputs and outputs): material (ingredients, trays, food waste), people (students in a queue), information (menus, orders, meal-plan verification), money (payments, supplier bills), energy (gas, electricity, hot water). Label every arrow with what moves: it shows what the next step needs.

The “Fix the Diagram” walkthrough (a lecture interactive, find at least four errors; answer key): error 1 — no start/end symbols → add Start and End; error 2 — “Kitchen” is a place, not a step → rename it “Prepare meals”; error 3 — the decision has only a “Yes” exit → add a “No” exit, e.g. “Prepare more meals”; error 4 — “Chop onions” is level-2 detail → keep one level: “Prepare meals”; error 5 — unlabeled arrows → label what moves: students, meals, money. The rules matter because the reader should never have to guess.

Level 0 of the cafeteria (worked example): inputs — food deliveries, staff arriving, students coming in; outputs — meals served, food waste, payments; context — the sanitary code, opening hours, the kitchen budget, the building layout; the inside is hidden. Level 1: Start → 1 Receive deliveries (input: food deliveries) → 2 Store ingredients (ingredients) → 3 Prepare meals (cooked meals) → 4 Serve meals (meals to students) → 5 Take payment → 6 Clear and dispose (used trays) → End. Compared with level 0: deliveries enter at step 1, students at step 4; payments leave at step 5, waste at step 6.

Level-1 PFD: University Cafeteria

Week 3 · steps are named “verb + noun”, every arrow is labeled

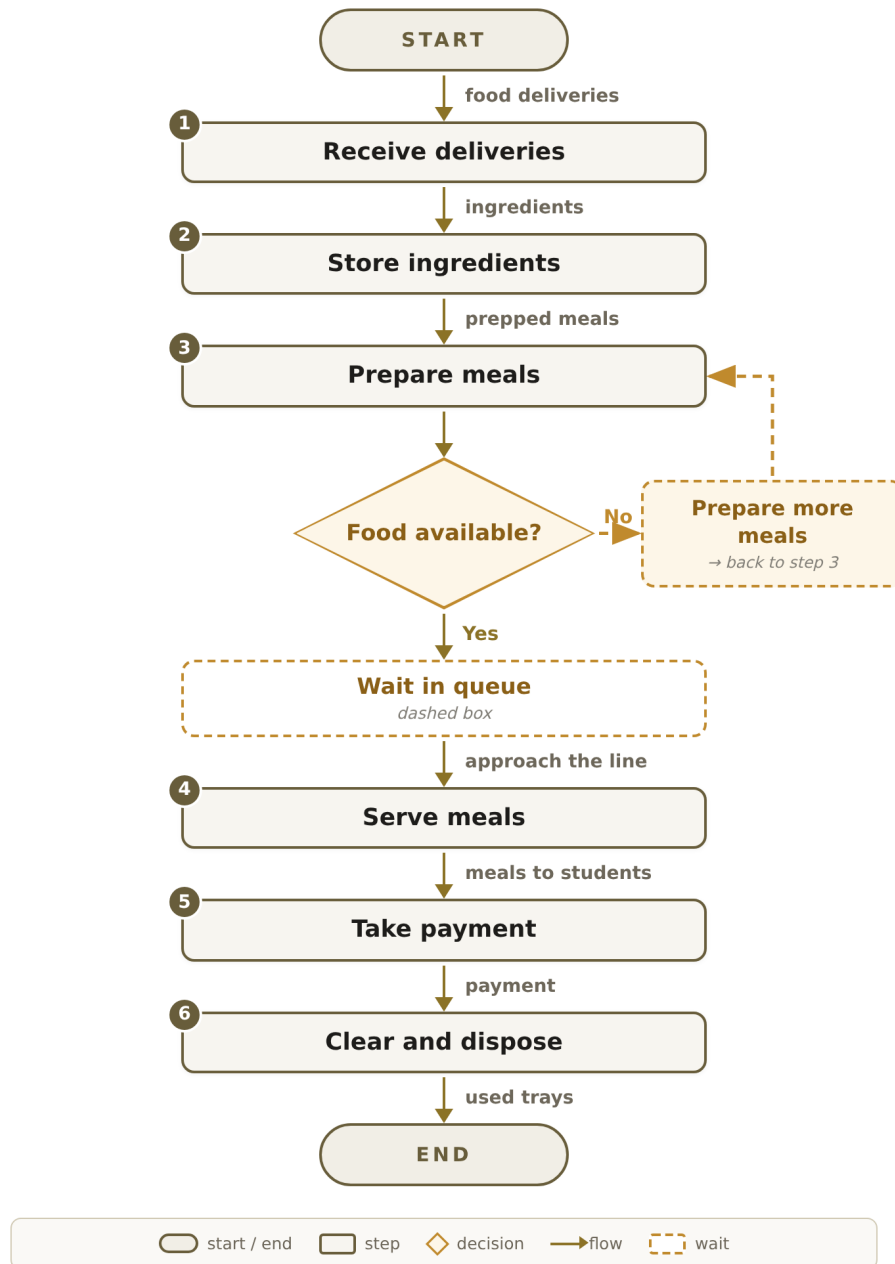


Fig. 4. Level-1 PFD of the university cafeteria (week 3): six steps, a decision with two labeled exits and the rework loop.

Decisions and waits: a dashed box marks a wait; a diamond marks a choice. Example: Wait in queue → Serve meal → Food available? — Yes → Take payment; No → Prepare more meals → back to step 3. “Queues and decisions are where systems slow down. We will measure them in week 9” (the dynamic behavior of systems: delays, queues, throughput, bottlenecks). The swimlane version (who does each step): the lanes Student / Kitchen staff / Cashier; the numbers show the order: 1 Join queue (student) → 2 Serve meal (kitchen) → 3 Receive meal (student) → 4 Take payment (cashier) → 5 Return tray (student). Every hand-off between lanes is a place where delays and errors hide.

A diagram is a model with a purpose (Harry Beck's example, the London Underground map of the early 1930s): it gives up real distances and street geography, but keeps the stations, the lines, and the interchanges. What it keeps: stations · lines · connections and interchanges; what it drops: real distances · street layout · surface geography. “Every diagram keeps what serves the purpose and drops the rest — that is what makes the model useful” (the idea returns in weeks 4–5).

One system — five perspectives (the bus route through five pairs of eyes): Rider — Walk to the stop → Wait → Board → Ride → Get off; success = a short wait, a low fare, safety. Driver/dispatcher — Start the shift → Check the bus → Run the route → Refuel → End the shift; success = on time, safe, a workable schedule. Maintenance crew — Inspect → Diagnose → Repair → Return to service; success = safe buses, planned downtime. University administration — Set the budget → Set the timetable → Review ridership → Adjust; success = cost per passenger, coverage, few complaints. Neighbor — Hear the bus → Notice the traffic → Contact the university; success = a quiet street, less traffic. Takeaways: different boundaries (the rider's picture starts at the stop, the driver's — at the depot), different steps (each includes what they see), different definitions of success (faster/safer/cheaper turn into competing definitions of a good job). “No single diagram is wrong. Each is a model built with a purpose. Good engineers draw several and ask: whose view is missing?”

Looking ahead — numbers on the arrows (weeks 6–7; the example was already worked through in lecture 3): community A ships 600 tons/day of waste and community B 400 tons/day to the Transfer Center; 25% is recycled, and the incinerator burns 200 tons/day. Nothing accumulates, so input = output: $1000 - (250 + 200) = 550$ tons/day to the landfill. Source: Kosky et al., Exploring Engineering, ch. 18, Example 18.2 — the full walkthrough is in section 4.3.

Self-study after lecture 3 (for yourself, not submitted): 1) draw the Level 0 of your system, then Level 1 with 5–8 steps; 2) label every arrow and note where the arrows cross the boundary; 3) redraw one part from a second stakeholder's point of view; 4) ask each stakeholder what “works well” for them. This is working material for the next step — turning observations into a defined problem.

4.2. Week-3 savollar (questions) — questions with model answers

Q: Warm-up: the supplier's truck and the billing office — inside or outside the cafeteria boundary? (carried over from lecture 2; a one-sentence defense of the answer is expected)

Answer: The truck: outside — deliveries cross the boundary as an input at step 1, and the truck itself is a supplying stakeholder. The meal-plan office: depends on the boundary — outside, if the diagram ends at “Take payment”; inside, if it includes “Verify meal plan”/“Update balance”. “There is no single right answer — there is a justified one”.

Q: Cafeteria elements and processes: list five nouns and five verbs.

Answer: Elements (nouns): cooks, ovens, trays, tables, students. Processes (verbs): receive, store, prepare, serve, pay, clear. Check: a process reads as “verb + noun” — prepare meals, not meals.

Q: The three levels of detail of the cafeteria: Level 0, Level 1, Level 2 — state each one.

Answer: Level 0 (black box): food, staff, and students come in — meals, waste, and payments go out; the inside is hidden. Level 1 (main steps): Receive → Store → Prepare → Serve → Pay → Clear. Level 2 (zoom in) on “Prepare meals”: Wash → Chop → Cook → Hold hot. Rule: one level per diagram; the level is chosen to fit the question.

Q: Fix the Diagram: find at least four errors in the diagram “Kitchen → Students enter → Food available? (Yes) → Chop onions → Serve meals → Pay”.

Answer: The full key (five): 1) no Start/End — add them; 2) “Kitchen” is a place, not a step — rename it “Prepare meals”; 3) the decision diamond has only a “Yes” exit — add “No: Prepare more meals”; 4) “Chop onions” is level 2 inside level 1 — replace it with “Prepare meals”; 5) unlabeled arrows — label them (students, meals, money). Other findings are possible.

Q: Which five categories of flow does a PFD arrow carry? Give an example of each for the cafeteria.

Answer: Material: ingredients, trays, waste. People: students in a queue. Information: menus, orders, meal-plan verification. Money: payments, supplier bills. Energy: gas, electricity, hot water. The arrow label shows what the next step needs.

Q: Why is Harry Beck's metro map a “model with a purpose”? What does it drop, and what does it keep?

Answer: It drops: real distances, street layout, surface geography. It keeps: stations, lines, interchanges. The purpose is to help a passenger plan a trip along lines and interchanges; distances are not needed for that. Lesson: a model is useful precisely because it drops the inessential; in weeks 4–5 the same idea returns in engineering models.

Q: Five stakeholders of the bus route: each has its own process and its own “good”. State the table.

Answer: Rider: Walk to stop → Wait → Board → Ride → Get off; success — a short, predictable wait, a low fare, safety. Driver/dispatcher: Start shift → Check bus → Run route → Refuel → End shift; success — a sustainable schedule. Maintenance: Inspect → Diagnose → Repair → Return to service; success — safe buses, planned downtime. Administration: Set budget → Set timetable → Review ridership → Adjust; success — cost per rider, coverage, few complaints. Neighbor: Hear bus → Notice traffic → Contact university; success — quiet, less traffic. Different starts, different steps, different ends — and all of it is one route.

Q: The “Three Lenses” exercise (10 minutes, in roles): draw a 5-step flow of the bus route as the Rider / Driver / Maintenance and compare.

Answer: Rider: Walk to the stop → Wait → Board → Ride → Get off. Driver: Start the shift → Check the bus → Run the route → Refuel → End the shift. Maintenance: Inspect → Diagnose → Repair → Return to service. Comparison: what is included/excluded, and where the diagrams conflict; the conflicts are competing definitions of success, which week 4 turns into different performance variables.

Q: Lab 3, small-group practice: what does the task include, and what is the deliverable?

Answer: The task (teams of 3–4, 15 minutes of work + 5 minutes of show-and-tell): define the system in one sentence (what it is and what it does); draw the boundary and declare what is inside/outside; list the inputs and outputs across the boundary; identify the stakeholders Uses It / Runs It / Affected By It; note the operating context; propose one concrete change and why. The deliverable: a Level 0 black box and a Level 1 box-language diagram on paper or a whiteboard — with a label on every arrow.

Q: Lab 3, the reference analysis of the “Solar Panel Assembly Line”: system, boundary, stakeholders, inputs/outputs.

Answer: The conditions: 600 glass sheets and 600 backsheets arrive daily, 36,000 solar cells, EVA film, aluminum frames, and junction boxes; the materials are inspected, the cells are sorted and strung, the modules are laid up with EVA and backsheets, laminated, and go into a 30-minute cooling queue; the panels are trimmed, framed, and fitted with boxes; every module is flash-tested and checked by electroluminescence, then a decision: pass / rework / scrap; approved modules are packed and shipped. Reference: the system — the assembly line from the receiving docks to the shipping docks; inside the boundary — inspection, stringing, lay-up, lamination, cooling, trimming/framing, testing, packing; outside — raw-material suppliers, the customer, the waste handler. Stakeholders: Uses It — the customer receiving the panels; Runs It — line operators, quality controllers, the shift supervisor; Affected By It — the maintenance crew, logistics, the environment (the scrap and waste streams). Inputs: 600 glass sheets/day, 600 backsheets/day, 36,000 cells/day, EVA, frames, boxes. Outputs: finished panels (shipment), modules sent for rework, scrap modules.

Q: Lab 3, Level 0 and Level 1 of the solar line + the decision rule.

Answer: Level 0: inputs — glass sheets (600/day), backsheets (600/day), cells (36,000/day), EVA/frames/boxes; outputs — finished panels, rework modules, scrap modules; context: the delivery schedule, the shop-floor capacity, the IEC quality standards, the 30-minute cooling requirement. Level 1 (box language): Start → 1 Receive materials (cells) → 2 Inspect materials (stack) → 3 Sort and string cells → 4 Lay up and laminate (modules) → 5 Cool, 30 min [wait] → 6 Trim, frame and fit boxes (panels) → 7 Flash test and inspect (approved) → 8 Pack modules → Ship to customer. The decision rule: both exits of every decision get a destination; rework returns upstream (to step 4); scrap leaves the system as waste. The reference answer stresses: “this is an approximate solution; it may vary depending on how the logic is understood”.

4.3. Week-3 savollar (questions) and formulas from the textbooks: mass balance (Kosky ch. 18)

Kosky, chapter 18 “Environmental Engineering” — the source of lecture 3 for “numbers on the arrows”. Environmental engineering is the application of engineering principles in order to (1) protect the health of people, (2) protect the ecosystems of nature, (3) improve the quality of life connected with the environment; until the mid-1960s it was called sanitary engineering within civil engineering. The field's tasks: air, water, and noise pollution control; the preparation and distribution of drinking water; the collection and disposal of

wastewater; the management of solid and hazardous waste; the cleanup of contaminated sites; environmental assessments and audits.

How pollution is measured (18.3): the amount of a pollutant is given by its concentration — mass per unit volume of the mixture. Formula (18.1): $C_A = m_A/V_{\text{mixture}} = m_A/(V_A + V_B + V_C)$. For water: 1 ppm = 1 mg/L and 1 ppb = 1 µg/L; “one cup of pollutant in a swimming pool is ppm, one drop in a swimming pool is ppb”; $1 \text{ g/m}^3 = 1 \text{ mg/L} = 1 \text{ ppm}$; $1 \text{ mg/m}^3 = 1 \text{ µg/L} = 1 \text{ ppb}$. For soil: 1 ppm = 1 mg/kg, 1 ppb = 1 µg/kg. For air: concentrations in mg/m^3 depend on temperature and pressure, so they are standardized at 1 atm (101,325 N/m²) and 25 °C (298 K); the molar volume $V_{\text{molar}} = RT/p = 8.3143 \times 298/101325 = 0.02445 \text{ m}^3/\text{mol} = 24.45 \text{ L/mol}$. Conversion (18.2a): concentration (mg/m^3) = concentration (ppm) $\times M/(24.45 \text{ L/mol})$; (18.2b): concentration (ppm) = concentration (mg/m^3) $\times 24.45/M$, where M is the molar mass of the gas. Example 18.1: CO₂ at 405 ppm in 2017 $\rightarrow 405 \times 44/24.45 = 729 \text{ mg/m}^3$.

The mass balance equation (18.3a) — the main tool: [mass accumulated in the control volume] = [total mass that entered] – [total mass that left] + [mass created or destroyed inside]. The rate form (18.3b): $(\dot{m})_{\text{CV}} = \Sigma(\dot{m})_{\text{in}} - \Sigma(\dot{m})_{\text{out}} + \dot{M} G$, where $\dot{m} = dm/dt$ is the mass flow rate, and $\dot{M} G$ is the generation rate inside the control volume due to reactions. Mass as such is neither created nor destroyed (except in nuclear reactions), but the mass of particular chemical species can be created/destroyed by reactions ($2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$).

Example 18.2 (cited by lecture 3): the Transfer Center receives 600.0 tons/day of solid waste from one community and 400.0 tons/day from another; 25% of the total is recyclables; the incinerator at full capacity burns 200.0 tons/day. How much waste should be planned for landfill disposal? Need — the rate of disposal to the landfill; Know — the mass flow rates at the input and output; How — define the system as the Transfer Center, draw a control volume with the inflows and outflows, apply the balance (18.3b); Solve: there are no reactions, so $\dot{M} G = 0$, and there is no accumulation, so $(\dot{m})_{\text{CV}} = 0 = \Sigma(\dot{m})_{\text{in}} - \Sigma(\dot{m})_{\text{out}}$; $0 = (600 + 400) - [0.250 \times (600 + 400) + 200 + \dot{m}_{\text{LF}}]$; $\dot{m}_{\text{LF}} = 550.0 \text{ tons/day}$.

Linking concentration and flow rate (18.4): $\dot{m}_p = C_p Q_f$, where $Q_f = dV/dt$ is the volumetric flow rate of the carrier medium (m^3/s), and C_p is the concentration of the pollutant. An alternative form of the balance (18.5): $(\dot{m}_p)_{\text{CV}} = 0 = \Sigma(C_p Q_f)_{\text{in}} - \Sigma(C_p Q_f)_{\text{out}} + \dot{M} G_p$. If the pollutant's volume is small relative to the water/air, the equation $C = 1.0$ (100%) is added at the inputs and outputs for the carrier medium. Example 18.3: a factory emits 50.0 kg/day of SO₂; a wind of 6.00 m/s drives $7.00 \times 10^6 \text{ m}^3/\text{s}$ of air with a background concentration of 0.050 µg/m^3 over the city — predict the SO₂ concentration downwind, behind the stacks (method: a control volume over the city, an SO₂ balance in the form (18.5)).

4.4. Week-3 atamalar (terms) — glossary

Term (EN)	Atama (UZ)	One-phrase definition
Process Flow Diagram (PFD)	Jarayon oqim diagrammasi (PFD)	Shows the steps of the process in order and what flows between them
Sequence	Ketma-ketlik	What happens first, next, and last

Term (EN)	Atama (UZ)	One-phrase definition
Flow	Oqim	What moves between the steps: material, people, information, money, energy
Black box (Level 0)	Qora quti (Level 0)	A diagram with only inputs and outputs; the inside is hidden
Main steps (Level 1)	Asosiy qadamlar (Level 1)	A handful of main steps inside the boundary
Zoom in (Level 2)	Yaqinlashtirish (Level 2)	One level-1 step expanded into its own diagram
Start / End	Boshlanish / Tugash	The symbols for the start and end of the process; one start, every path reaches an end
Step	Qadam	A single action, named “verb + noun”
Decision	Qaror	A diamond question with labeled Yes/No exits, both leading somewhere
Wait / queue	Kutish / navbat	A dashed box where things sit and wait; queues slow systems down
Swimlanes	Yo'laklar (swimlanes)	Splitting the diagram by performer; hand-offs between lanes hide delays
Control volume	Nazorat hajmi	The defined region of space for which the mass balance is written
Mass balance	Massa balansi	Accumulation = input – output + generation; with no accumulation and no reactions, input = output
Concentration	Konsentratsiya	Mass of pollutant per unit volume of the mixture (ppm, mg/L, mg/m ³)
Molar volume	Molyar hajm	24.45 L/mol for an ideal gas at 1 atm and 25 °C
Perspective	Perspektiva	A stakeholder's view: its own boundary, its own steps, its own definition of success

Table 4.1. Week-3 terms (English — Uzbek)

5. Week 4 — Problem Framing and Modelling I

5.1. Key points (“all qoidalar” of week 4)

Week 3 recap (in one breath): systems are interacting elements and processes that turn inputs into outputs; PFDs show the steps inside the boundary and what flows between them; perspectives — different stakeholders draw different diagrams of the same system; success — every stakeholder also defines “works well” differently. Today: how to turn those views into a clear problem and measurable variables.

The lecture's opening dilemma (“What Is the Problem?”, an adaptation of the Empire State Building case from Kosky ch. 23): a fictional client reports that workers in a tall office building complain about long

elevator waits. Option A: “Design a new elevator for the building.” Option B: “Increase employee satisfaction with the elevators in the building.” Which one leaves more options open? The right choice is B: formulation A sews the solution into the problem itself, whereas B admits mirrors by the elevators, free coffee on the busiest floors, a new schedule — and only then a new elevator. The lecture's warning about “problems with problems”: problems tend to change our behaviour and make us do what we instinctively know is wrong; different types of problems call for different tactics.

Four kinds of information (the Kosky ch. 23 classification, “Clarifying the Problem”): **Fact** — something observed or measured (“the lunch queue reaches the door at 12:40 on Monday — counted”). **Inference** — a conclusion drawn from facts (“queues are longest when two class periods end at the same time”). **Speculation** — a guess without evidence so far (“maybe the cashier is the slowest link”). **Opinion** — a personal view or preference (“the cafeteria should open a second serving line”). All four are useful: opinions and guesses show what people care about; facts tell you what is happening. But only facts and good inferences may anchor a problem statement.

The three-step check of what you hear (Test What You Hear): 1) Hear — collect what people say, including opinions and guesses; 2) Observe and measure — watch the process; count, time, ask stakeholders directly; 3) Decide — keep what the facts support; reframe what they contradict. Example: many shoppers say the store needs more cash registers; watching the queue all day and talking with the manager may confirm this — or reveal a different cause. Your Week 3 PFD tells you where to look: at the steps, the waits and the hand-offs.

Walk-through of the “Sort the Statements” interactive (full key): A “Lunch queues exceeded 15 people on 3 of 5 days this week” (counted by a student) — **fact**, counted with a time period. B “The cafeteria is badly designed” — **opinion**, a judgement without evidence. C “Queues are longer on the days soup is served” — **inference**, a pattern read out of observations. D “I think the cashier is too slow” — **opinion**, a personal view. E “The kitchen probably runs out of food early” — **speculation**, a guess about a cause, not verified. F “The average wait was 9 minutes at 12:30 on Tuesday” (timed with a phone) — **fact**, measured with a time and a method.

From complaint to problem statement (Complaint → Need → Problem Statement): a complaint describes the symptom (“We wait too long for the elevators”); a need describes what the person actually requires (“Employees need to get to their floors without exhausting delays”); a problem statement turns the need into what we work on (“Increase employee satisfaction with the elevators”). The course template for a problem statement: **[Stakeholder] needs [outcome] in [system], judged by [measure], within [constraints]**. Example: “Students need short, predictable lunch waits in the cafeteria, judged by the peak-time waiting time, within the current kitchen budget.”

The solution-free test — the week's key rule: does the problem statement name a device, an app, a vehicle or a building? If yes — rewrite it. “One word can hide a solution”: the hidden solution often lives in a single noun — “Design a new elevator” (a new elevator; it rules out scheduling, mirrors, coffee); “Design a device to eliminate the blind spot” (a device; it rules out simply rearranging the mirrors); “Design a blimp to retrieve the ball” (a blimp; it rules out the extendable arm — the Blimp Wars case from Kosky). Once a noun is in the statement, the other answers are no longer considered.

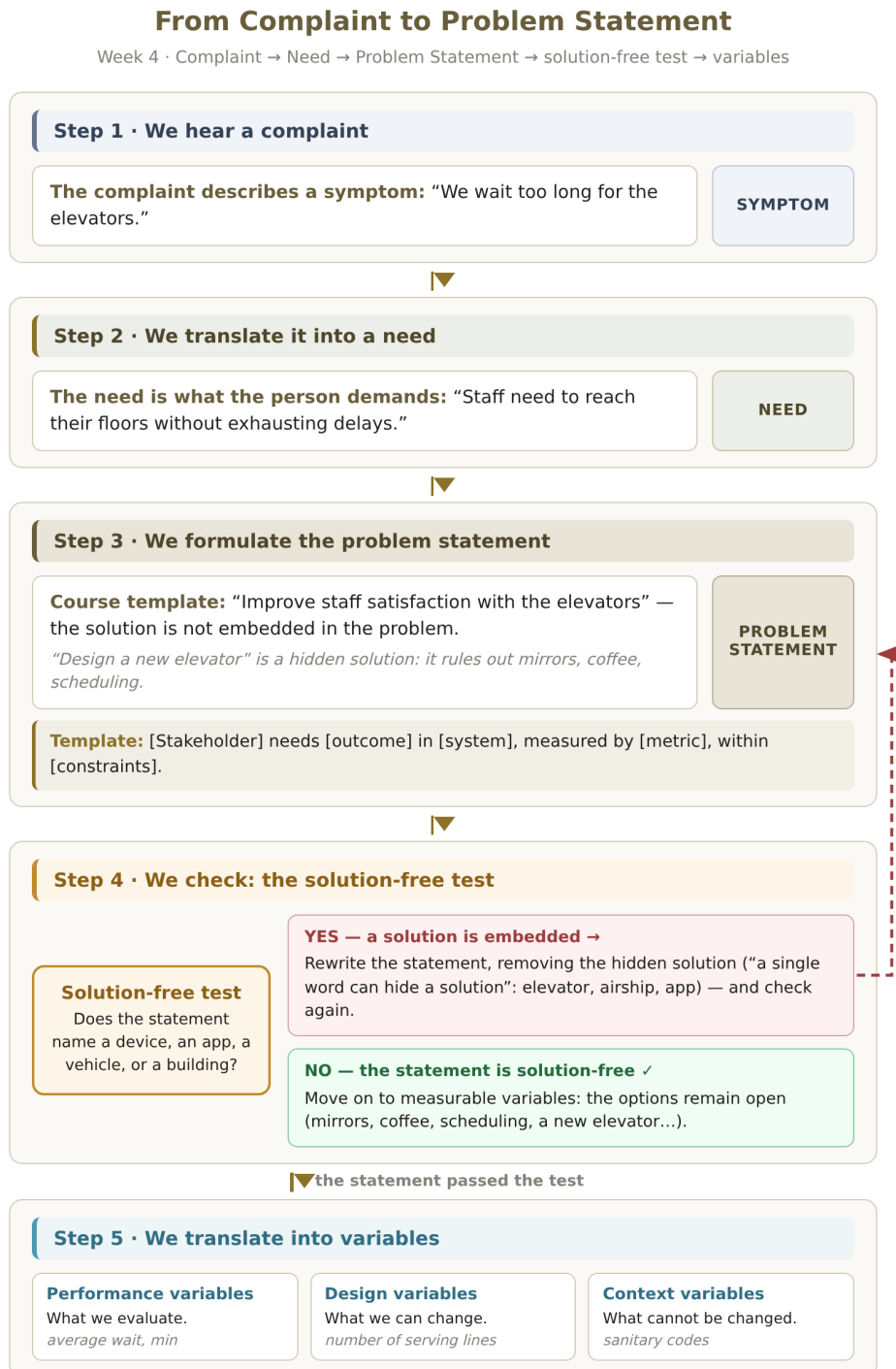


Fig. 5. From complaint to problem statement (week 4): Complaint → Need → Problem Statement, the solution-free test and three types of variables.

Reference rewrites (the “Rewrite the Problem” interactive, sample answers): 1) “Build a bigger cafeteria” — hidden solution “a bigger building”; rewrite: “Students need to find a place and eat without long delays, judged by the time from entering to being seated at peak.” 2) “Develop an app showing bus arrival times” — hidden solution “an app”; rewrite: “Passengers need to know when the next bus will arrive, judged by the share of passengers able to plan their wait.” 3) “Install more washing machines in the dormitory laundry” — hidden solution “more machines”; rewrite: “Residents need to wash their clothes without long waits, judged

by the time from arriving to starting the wash.” The check is the solution-free test, not matching the exact wording.

An engineering variable: a variable is a quantity that can be measured, expressed as a number with units. Time is measured with a clock (seconds, minutes), length with a ruler (metres), temperature with a thermometer ($^{\circ}\text{C}$ or kelvins), counts with “how many students, meals, buses”. Speed, for example, combines a ruler measurement (distance) with a clock measurement (time). A number without units is incomplete: “Waiting time: 12” — 12 seconds? 12 minutes? A student at the front of the queue and a student at the back would read it very differently; “Speed limit: 100” — 100 km/h $\approx$ 62 mph: one number, two different speeds. In engineering a result has two parts — the value and the units — and both must be right (Kosky ch. 2).

Variables live on the diagram: every step and arrow of the Week 3 PFD suggests what can be counted, timed or weighed. Cafeteria: 1 Receive deliveries — deliveries per day (count/day); 2 Prepare meals — meals prepared per hour; 3 Serve meals — waiting time at the line (min); 4 Take payment — payments per minute (count/min); 5 Clear and dispose — food waste (kg/day).

Three kinds of variables: **performance variables** — what we judge (average waiting time, min; meals per hour; waste, kg/day); **design variables** — what we can change (the number of serving lines, staff per shift, menu size); **context variables** — what we cannot change (opening hours, sanitation codes, building size). Performance variables say whether the system works; design variables are the levers; context variables are the limits.

From vague words to variables (the lecture's table): “crowded” — people in the queue (count or people/m²); “slow” — waiting time at the line (minutes); “wasteful” — discarded food (kg/day); “expensive” — cost per meal (currency/meal); “unreliable” — late deliveries per week (count/week). A useful variable answers “how many?” or “how much?”. “If you cannot picture how to measure it — keep looking.”

Stakeholders suggest the measures (a bus route, the same five perspectives as Week 3): Rider — a short, predictable wait — waiting time at the stop (minutes); Driver/dispatcher — a schedule that can be kept — trips completed on time (% of trips); Maintenance — safe, reliable buses — breakdowns per month (count/month); Administration — good use of the budget — cost per passenger (currency/passenger); Neighbor — less noise and traffic — buses per hour on the street (buses/hour). Each perspective points to its own performance variable; the values are illustrative, for discussion.

Demands (demand) and wishes (wish): **D (demand)** — must be met; without it the design is unacceptable (example: every bus passes a safety inspection before entering service). **W (wish)** — nice to have; desirable but not critical for success (example: a real-time display at every stop). Use numbers wherever possible, even rough ones: “average peak wait under 10 minutes”, not “short waits”. The numbers are refined later. Do not turn a wish into a demand — it makes the design harder than it needs to be. Source: Kosky ch. 23, “List of Design Specifications”.

Four signs of a good performance measure: 1) Measurable — you can count, time or weigh it, with a declared method. 2) Relevant — a real stakeholder cares about it. 3) Solution-free — it does not presume how the problem will be solved. 4) Specific — it has units and a target, even a rough one. Beware of measures that hide trouble: the average waiting time can look fine while some passengers wait a very long time.

A working definition of system performance (the course template): **The performance of [system] is judged by [variable 1], [variable 2] and [variable 3], each with units and a target, through the eyes of [stakeholders].** Example (cafeteria, illustrative): performance is judged by the average peak-time wait (min), meals per hour (count/h) and food waste (kg/day), through the eyes of the diners, the kitchen staff and the administration. “Working definition” means it can change as you learn; write it down so the team agrees on it.

The Need-Know-How-Solve method (Kosky ch. 2 — a “fail-safe” method for setting up and solving problems): **Need** — the variable (or variables) we are looking for; written down first. **Know** — the known quantities: from the problem statement or from background knowledge; they may be numbers, sketches, diagrams, principles. **How** — the solution method, usually an equation, including the assumptions; written down before it is applied. **Solve** — only now do we compute, check the units, interpret. The method: 1) leaves an “audit trail” of the solution; 2) formalises and simplifies the problem; 3) protects against the temptation to “glance and write down the answer”; 4) earns partial credit even if the answer is wrong. A clear problem statement and well-chosen variables give you the Need; the next step (Week 5) is How: models as simplified representations built on assumptions.

The Need — Know — How — Solve Method

Kosky, ch. 2 · a “fail-safe” method for setting up and solving engineering problems

NEED What we look for — the unknown variable

The variable (or variables) that must be found; written down first — always together with units.

Example: X — the length of the straight segment, miles; t — the waiting time, minutes.



KNOW What is known — the given data

Known quantities from the problem statement or background knowledge: numbers, sketches, diagrams, principles.

Example: a 2-mile segment at 45°; a queue of 15 people; a limit of 100 km/h $\approx$ 62 mph.



HOW How we solve it — the method before the computation

The equation or solution method, including assumptions; written down before any numbers are substituted.

Example: the Pythagorean theorem $x^2 = 2^2 + 1^2$; mass balance: $\text{accumulation} = \text{input} - \text{output}$.



SOLVE We compute — and check the result

Substitution, arithmetic, checks of units and significant figures, interpretation of the answer.

Example: $X = 1.6$ miles; “12 minutes”, not just “12”.

What the method gives you: leaves an “audit trail” of the solution · formalizes and simplifies the problem · protects against the temptation to “glance and jot down an answer” · earns partial credit even if the final answer is wrong.

Fig. 6. The Need-Know-How-Solve method (Kosky, ch. 2): the four steps of solving an engineering problem.

Week 4 independent work (for the team, not submitted): 1) write a one-paragraph problem statement for your system — in terms of needs, not solutions; 2) list 3–5 performance variables, each with a unit and a way of measuring it; 3) label each one as a demand or a wish and give a rough numerical target; 4) check the statement and the variables against the solution-free test.

5.2. Week-4 savollar (questions) — questions with model answers

Q: The elevator dilemma: A “Design a new elevator” or B “Increase employee satisfaction with the elevators”? Which option leaves more options open, and why?

Answer: B. A sews the solution (a new elevator) into the statement — whereas the complaint was about waiting; B admits mirrors by the elevator doors, free coffee on the busiest floors, rescheduling, and only then a new elevator. Kosky's evidence (ch. 23): the improved Empire State Building formulation — “Increase customer satisfaction with the elevators” — is exactly how the real case was solved.

Q: Classify the six statements (Sort the Statements) and explain why.

Answer: A — fact (counted, with a time period); B — opinion (a judgement without evidence); C — inference (a pattern from observations); D — opinion (a personal view of the cashier); E — speculation (a guess about a cause, not verified); F — fact (measured with a time and a method). To remember: facts and good inferences may anchor a problem statement; opinions and speculations are material for interviews and observations.

Q: The three steps of “test what you hear” and their connection to the PFD.

Answer: 1) Hear (collect everything, including opinions); 2) Observe and measure (count, time, ask directly); 3) Decide (keep what the facts support, reframe what they contradict). Connection: the Week 3 flow diagram shows where to observe — the steps, the waits, the hand-offs.

Q: Build a Complaint → Need → Problem Statement for: “drivers think the charging station is slower than its competitors” (the Lab 4 case).

Answer: Complaint: “Drivers think the station is slower than its competitors.” Need: “Customers need to charge and check battery health without a long wait.” Statement (per the template): “Customers need a short, predictable total time at the station, judged by the time from arrival to release, within the grid operator's peak-power limit.”

Q: Find the hidden solution and rewrite without it: (1) “Build a bigger cafeteria”; (2) “Develop an app with bus times”; (3) “Install more washing machines”.

Answer: (1) hides “a bigger building”; rewrite: students need to find a place and eat without long delays, judged by the time from entering to being seated at peak. (2) hides “an app”; rewrite: passengers need to know when the next bus will arrive, judged by the share of passengers able to plan their wait. (3) hides “more machines”; rewrite: residents need to do their laundry without long waits, judged by the time from arriving to starting the wash. The check is the solution-free test: no device, no app and no building is named.

Q: What is an engineering variable, and why is “Waiting time: 12” an incomplete answer?

Answer: A variable is a measurable quantity expressed as a number with units. “12” without units is incomplete: 12 seconds versus 12 minutes are different systems; an answer has two mandatory parts — the value and the units. In engineering a result must be right in four ways: the right variables, the right numbers, the right units, the right number of significant figures (Kosky, the ch. 2 summary).

Q: Three kinds of cafeteria variables — one of each kind.

Answer: Performance (what we judge): the average waiting time (min). Design (what we can change): the number of serving lines. Context (what we cannot change): the sanitation code/opening hours. The lecture's mnemonic: performance — “what we judge”, design — “the levers”, context — “the limits”.

Q: The “Find the Variables” assignment (a bus route): 5 variables with units, at least 2 performance, 1 design, 1 context, and a way of measuring each.

Answer: Model set: performance 1 — waiting time at the stop (min; time it with a phone at the peak stops); performance 2 — cost per passenger (currency/passenger; budget / number of passengers counted by the fare validators); design 1 — the bus headway (min; the depot's schedule); design 2 (extra) — the number of buses on the line (count); context 1 — the speed limit on the route's streets (km/h; the traffic code). Measuring: what instrument, who and when — for example, “waiting time: a student observer times it from arriving at the stop to boarding, 5 weekdays, 7:30–9:00”.

Q: Stakeholders Suggest the Measures: reduce the five perspectives of the bus route to a table “stakeholder — success — variable — unit”.

Answer: Rider — a short, predictable wait — waiting time at the stop — minutes. Driver/dispatcher — a schedule that can be kept — % of trips on time — %. Maintenance — safe, reliable buses — breakdowns per month — count/month. Administration — good use of the budget — cost per passenger — currency/rider. Neighbor — less noise and traffic — buses per hour on the street — buses/hour. Each Week 3 perspective points to its own Week 4 performance variable.

Q: Demand or wish? (1) every bus passes a safety inspection; (2) a real-time display at every stop; (3) an average peak wait under 10 minutes.

Answer: (1) — a demand: without the safety inspection the design is unacceptable. (2) — a wish: nice to have, not critical; turning a wish into a demand makes the design harder than justified. (3) — a numerical target; if it is a condition of acceptability, it is a demand with a number, otherwise a wish; the week's own rule: “numbers wherever possible, even rough ones”.

Q: The four signs of a good performance measure and the trap of averages.

Answer: 1) Measurable (the method is declared); 2) relevant (a stakeholder cares); 3) solution-free (it does not presume a solution); 4) specific (units + target). The trap: the average waiting time can look decent while some passengers wait a very long time — so look at the distribution, not only at the average.

Q: The Need-Know-How-Solve method: name the steps and the connection to Week 4.

Answer: Need — the variable we seek (written down first); Know — what is known (from the problem statement and from knowledge; numbers, sketches, principles); How — the method/equation + assumptions (before computing); Solve — we compute, check the units, interpret. Connection: a clear problem statement and chosen variables give the Need; the method protects against “blind number plugging” and leaves an audit trail; Kosky's examples: the Texan's ranch (a quadratic equation, $X = 1.6$ miles), the Schenectady barbershops (an estimate of 21 versus 23 in the directory, ~10% accuracy).

Q: Lab 4, Part A “Classify and Frame” for the EV Charging and Battery Test Station card.

Answer: Facts of the card: cars arrive at an average of 12 per hour; 4 charging bays; charging 30–40 min + a 10-minute diagnostic; an inspection when battery health is below 80%. Sorting: fact — arrivals of 12 cars/hour and 4 bays (counted); inference — queues form when arrivals outrun the combined charging+diagnostic time of the four bays; speculation — the station may be slower because of conservative safety limits; opinion — customers want to wait less; the grid operator worries about peak power. Complaint → Need → Problem Statement: see the model above (a short, predictable time from arrival to release, within the peak-power limit).

Q: Lab 4, Part B “Map the Process” for the EV station: 7 steps and one variable per step.

Answer: 1) Arrive — the arrival rate, cars/h; 2) Queue [wait] — waiting time, min; 3) Assign charger — bays occupied, count; 4) Charge (30–40 min) — charging time, min; the step variable: power per charger, kW (the grid operator's interest); 5) Diagnostic test — test time, min; the variable: the share of batteries below the threshold, %; 6) Check: Health below 80%? — battery health, %; 7) Inspect / Release — cars released, count/h; the key process variable: the total time at the station, minutes. 7 steps — within the 6–7 limit; the diamond leads to a technician inspection when health is below 80%.

Q: Lab 4, Part C “Stakeholders and Targets” for the EV station: the table and the targets.

Answer: Customer — a short, predictable visit — total time at the station — minutes. Station operator — high throughput at manageable demand — cars per hour — count/hour. Grid operator — low peak power — peak consumption — kW. Technician — dangerous batteries caught early — batteries flagged for inspection — % of tests. Targets: must-have (demand): the average total time at the station ≤ 45 min at peak; the peak power within the contracted limit (≤ 150 kW). Nice-to-have (wish): the average time ≤ 30 min — faster than the competitors; a real-time queue-length display. “Another well-justified set of variables and targets is graded just as highly.”

5.3. Week-4 savollar (questions) and qoidalar (rules) from the textbooks

Kosky, chapter 21 “Introduction to Engineering Design” — design and the systematic approach. The nature of engineering design: design creates new products, using the available technology to improve performance, reduce the price or reduce the risk; a bridge that is “visually stunning without a strength calculation” is design without engineering; a concept car evaluated by an analysis of aerodynamics, strength and manufacturability

alongside style is engineering design. Until the 1960s this description was sufficient; today the definition has been extended by a systematic thinking process and best practices (Design for Six Sigma). Why be systematic: designers search for the best solution under tight constraints of time and resources; the engineering approach minimises the number of design iterations — fewer iterations = cheaper and faster.

Design problems versus homework problems: a homework problem has a single unique answer, found by a logical method (need-know-how-solve); a design problem has no unique answer — several designs may satisfy the statement; there may be a “best” answer with respect to the critical requirements (a design that minimises unit cost may be unreliable — if reliability is critical to the client, it is not the “best” answer). The qualities of a good designer (Kosky 21.5): curiosity about how things work (take devices apart — reverse engineering; stroll through a toy store); selflessness (suppress your ego for the sake of the team); fearlessness (step into the unknown); persistence (failures are inevitable); adaptability (“if the ship is sinking, don't go down with it — redesign it”).

The design notebook (design notebook, 21.6) and the Gordon Gould case: a permanently bound notebook with numbered pages; on the night of November 9, 1957, Columbia University graduate student Gould wrote down and sketched the idea of amplifying light — “light amplification by stimulated emission of radiation” (laser) — and on the Wednesday had the notes notarised by a notary in a candy store; for 30 years he litigated patents against rivals whose filings came later, and won millions in royalties thanks to a notarised, clear and contemporaneous record. Rules for keeping the notebook: date and number every page; do not tear out pages; do not leave blank pages (cross them out with a diagonal); enter all data, descriptions, sketches, calculations; an index on the first page; write in real time, without rewriting a fair copy; write in pen; do not use correction fluid — strike through; paste in computer printouts, graphs, photographs; write as if someone else will read it; document the team meetings (date, results, tasks).

The eight steps of the engineering design process (Kosky 21.8, Fig. 21.1 — also a PFD!): 1) Define the problem. 2) Generate alternative concepts. 3) Evaluate and select a concept. 4) Detail the design. 5) Defend the design (design defense). 6) Manufacture and test. 7) Evaluate the performance. 8) Prepare the final report. The two goals of the systematic approach: to eliminate personal biases and to maximise the thinking and information gathering done before the final design is fixed. The “Tower” exercise: the tallest tower out of 5 sheets of paper in 10 minutes; the materials are distributed unevenly (tape, reinforced tape, paper clips, scissors — two teams per material); 1 point per inch of height; a 10-point bonus for the first team to finish; after 3 minutes, one person on each team is offered 8 points to switch to another team; the debrief: the quality of teaming between and within teams, the ethics of switching, the importance of the materials (manufacturability).

Kosky, chapter 22 “Design Teams”: “All teams are groups of people, but not all groups are teams.” Team members work toward a common goal; group members merely share common interests. Managing the project: understand the project's goals (be able to state them in one sentence); involve everyone (break the work into individual assignments); keep it moving (work methodically toward the milestones). The Gantt chart (1917, Henry L. Gantt) — horizontal task bars on the project's time scale; PERT — a network diagram with a critical path. The qualities of a team leader: the ability to delegate (know people's strengths and trust them), communication (state clearly what is to be done), a sense of humour (morale affects productivity),

confidence (the team takes its cues from the leader), engagement (lead by example), a positive attitude, creativity. Team self-assessment: a 10-question questionnaire (a clear vision of the goals; skills and resources; clear roles; time/place/resources; attendance; working together; understanding the requirements and the deadlines; communicating changes; the effectiveness of meetings; everyone's participation), on a scale Disagree = 0 / Undecided = 1 / Agree = 2, maximum 20; a low score is a signal for corrective actions.

Kosky, chapter 23 “Design Step 1: Defining the Problem” — the core of Week 4. Design begins when a client expresses a need; the engineer's job is to translate the client's needs into engineering terms: understanding the need, defining the problem and the list of design specifications. Need identification: the failure to identify, understand and validate the need BEFORE designing is one of the most common causes of failure of the whole design process; methods: interview potential clients, observe behaviour, keep a “bug list” of the irritating aspects of products; then research the market (consumer analysis through surveys), the competitors (competitor analysis), realism (“I want to go to Mars next week” — unrealistic), financial and legal risks (patents), profitability.

Defining the problem (23.3): the original definition “Design a new elevator for the Empire State Building” is biased; the better version — “Increase customer satisfaction with the elevators in the Empire State Building” — admits a mirror on the elevator doors or free coffee on the busiest floors. “Design a device to eliminate the blind spot in a vehicle” — the word device rules out the solution of simply rearranging the mirrors; “Blimp Wars” — the word blimp steered the students toward blimps instead of the extendable arm. Design specifications (23.4): the specification list “gives the client a voice”; the best way is a personal interview, recording the client's exact wording; classify each item as a demand (demand, D) or a wish (wish, W); do not confuse them: a wish treated as a demand complicates the design; use numbers wherever possible (“weigh less than 10 pounds”, not “low weight”); the specifications must be independent of the solution (“the wheels must be rubber” is biased twice: the wheels and the material). The categories of specifications: Performance, Geometry, Materials, Energy, Time, Cost, Manufacture, Standards, Safety, Transport, Ergonomics, Environment. Example 23.1 (the RC golf device): a complete list of D specifications across all 12 categories (a drive distance of 15–250 yards, a slope of up to 45°, 8 servos, a single housing, within a 3-foot circle, less than \$600, less than 14 weeks, FAA standards, a safety review, portability).

Clarifying the problem (23.5, “Clarifying the Problem”): start with what you know — people write down what they know, brainstorming generates the maximum; decide what information is missing (how many people use the bridge? when — all the time or in the morning and the evening? then the formulation: “Design a four-lane highway bridge for 400 cars per hour at peak”); gather the information — it falls into four categories: facts (15% of people do not use the bridge at all), inferences (a significant share of drivers bypass the bridge), speculations (lots of trucks), opinions (“I think the new bridge is too expensive”); speculations and opinions are especially important for gauging public opinion; then — define the problem in terms of needs, not solutions: by defining it through possible solutions you close the door on other, possibly more effective solutions. Milestone #1 (23.6): for a general project — interview the client and prepare a complete list of engineering specifications; for competitions — study the rules, supplement the list of requirements, look for “holes” in the rules, and do not bias the requirements toward a particular solution.

Kosky ch. 23, the Week 4 exercises (Exercises): 1) develop an engineering problem statement for the need “students need parking near campus”; 2) “shoppers need faster checkout in stores”; 3) “women need a place

for a handbag in a car”; 4) a list of desired (D) specifications for “a backpack that comfortably holds 10 kg of books, a laptop and lunch”; 5) for “an interactive toy for a blind child”; 6) for “a device that launches peanut-butter sandwiches at a crowd of students”; 7–9) five clarifying questions for each of problems 4–6 (for whom? under what conditions? what sizes/weights/materials? what budget? what safety concerns?); 10) from your own experience, propose a “need” of small children, and create a problem statement, a list of specifications and questions for the target group.

Dym, chapter 1 “Engineering Design” — the vocabulary of design. Three roles in any project: the client (client — the one who wants the design to be conceived), the user (user — the one who will use it), the designer (designer). Dym's formal definition: **engineering design is a systematic, intellectual process in which engineers generate, evaluate and specify solutions for devices, systems or processes whose forms and functions achieve the clients' objectives and the users' needs while satisfying the specified constraints**. The vocabulary: a design objective (design objective) — a property or behaviour we want; a constraint (constraint) — a limit/bound on properties or behaviour; violating it makes the design unacceptable (binary: yes/no, no intermediate states); a function (function) — what the device must do; engineering functions almost always involve the transformation or transmission of energy, information or material; a means (means) — a way to accomplish a function (friction is a means of braking force); form (form) — shape and structure as opposed to material. Objectives may be met fully, partially or not met; constraints must always be satisfied — hence the difference between “as cheap as possible” (an objective) and “cheaper than \$20” (a constraint).

Metrics and specifications (Dym 1.2.3): a metric (metric) — the scale on which achievement of an objective is measured; a specification (specification) — the scale on which the performance of functions is measured; the terms requirements and specifications are often synonymous, but Dym separates them: requirements — the client's requirements (objectives, constraints and functions in the client's words), specifications — engineering specifications (the designer's expression in engineering terms). Form and function are independent: from a smartphone's form you cannot infer its purpose, and knowing the function “wireless telephony” does not suggest the form. All design is systematic design (Herbert Simon: “design is the activity of intending to produce a description of an artifact in terms of its organization and functioning — its interface between the inner and outer environments”); the future challenges — complex systems (the interstate highways, the power grid, the internet). Communication: the product of design is usually a set of fabrication specifications, which separates “designing” from “manufacturing”; the Hyatt Regency case (Kansas City, 1981): the contractor could not obtain 24-foot threaded rods and hung the second-floor walkway from the fourth-floor walkway — the fourth-floor walkway's load was not designed for the doubled load; 114 people died and the losses ran into millions; the lesson — the specifications must be complete and unambiguous, conveying the designer's intent; design is a human, social activity.

Design problems are ill-structured and open-ended (Dym 1.3): ill-structured — solutions are not found by the routine application of formulas; in early design the formulas are unavailable or inapplicable; open-ended — several acceptable solutions exist; uniqueness, so important in mathematics, does not apply to design; the ladder example: the type of ladder depends on the uses; even having chosen a stepladder — wood, aluminium, plastic, composite? the price? the load? “can we determine the best or the optimal ladder? No.” Learning by doing: the studio aspect; designers, like dancers and athletes, use drills, coaches and the observation of masters. Project management — the 3S model: scope (what the project is to achieve),

schedule (resources available and used on time to an agreed date), spending (using only the necessary resources); tracking these three is project management.

Dym, chapter 3 “Problem Definition: Detailing Customer Requirements”: the client's initial statement may contain errors (wrong information, incomplete data — “Foothill Avenue” instead of boulevard), biases (bias — the problem may not be the interchange at all, but that the students cross the road in the wrong place) and implied solutions (“bottle” rules out the pouch and the tetra pak; “device”, when a process would be more apt). A quick reworking of the statement — eliminate the errors, the biases and the implied solutions. Framing the customer requirements: the questions “what properties/behaviours do you want?”, “what do you want it to do?”, “is there anything like it on the market?” plus “what do you mean by that? how would you do it? why do you need it? what should be avoided?”. The list of attributes is sorted into O (objectives — usually adjectives), C (constraints), F (functions — usually “verb-noun”), M (means — crossed out for the time being, because they are tied to a solution). The revised problem statement is the public face of the project: an important tool for communicating with the client and part of the team charter; be careful with the wording (“after consultations with the client we adopted the following statement”, not “having corrected the client's errors and biases”). The Danbury School case: a device to support the arm of a third-grader with cerebral palsy (CP) for drawing; Team A focused on drawing and added the constraint “set up in 8 minutes or less”, Team B took a wider set of activities (drawing, writing, eating) and “2–3 minutes of setup” — two teams, one client, two understandings of one problem.

Dym, chapter 4 “Clarifying the Objectives”: objectives are organised into a hierarchy (objectives tree) — “why?” questions move up (“a useful ladder” → “a marketable one”), “how?” questions move down; when the “why” question turns into a “how” — functions appear. Measuring: to rank the objectives — the pairwise comparison chart, where each objective is compared with every other and the winner of a pair gets a point; the sum of the points shows the priority. Metrics (4.4): a good metric actually measures the objective (the number of colours on the packaging is a bad metric of attractiveness); surrogate metrics (surrogate metrics) — when there is no direct ruler: dropping a phone from different heights for strength, the number of parts for simplicity, the assembly time; the units and the scale must fit (do not measure a ladder's weight in tonnes or in milligrams); points over the range (figures of merit: speed 50–200 km/h → 0–10 points, 170 km/h → 8 points); the Use-Value Analysis scale (0 absolutely useless ... 10 excellent) and the German standard VDI 2225 (0 unsatisfactory ... 4 very good/ideal) align the different metrics onto a common ruler; all objectives are measured consistently so that none dominates by its scale; if the measurement is too expensive — convert the objective into a constraint (for example, “contains no parts costing more than \$20”).

Dym, chapter 5 “Identifying Constraints”: a constraint is a limit or a prohibition on properties/behaviour; usually a binary choice (the material is a conductor or not; the deflection of a step is less than 0.05 inches or not); the alternatives that violate it are unacceptable. Constraints narrow the design space, objectives allow exploring what remains; a constraint is “weeding out”, an objective is “choosing among the acceptable”. Satisficing (Herbert Simon): designs that satisfy all the constraints but are not necessarily optimal — “satisfice”; one can minimally satisfy OSHA or significantly exceed it (a “super-safe ladder”) for the sake of marketing. Example: the objective “an inexpensive ladder” + the constraint “no more than \$25” — first the weeding-out by the constraint, then the choice by the price and the other objectives. The constraints for the juice container: chemically inert, no sharp edges; they are connected with the objectives safe and long shelf

life. For the Danbury project, Teams A and B found different values of the same constraint (setup ≤ 8 min versus 2–3 min) — perhaps because the teachers answered the requests to “give hard numbers” differently; or perhaps the short setup was an objective, not a constraint.

Dym, chapter 6 “Establishing Functions”: a function is what the device must do; a systems view helps — to do something = to transform an input into an output ($y = f(x)$). A bookcase “does nothing, it just stands there” — that is untrue: it resists the force of gravity in order to hold the books, and it allows organising them on shelves. Misunderstanding all the functions can be tragic (Hyatt Regency). The tools for establishing the functions: 1) Black box $\rightarrow$ glass box: draw the system with the inputs on the left and the outputs on the right (a drill: inputs — electrical power, a supporting force, speed/direction control; output — a rotating chuck), then “lift the lid” and expose the subfunctions (the cord transmits the energy $\rightarrow$ the switch directs and transmits information $\rightarrow$ the universal motor turns electricity into mechanics $\rightarrow$ the transmission steps up the torque, reducing the rotor's 30,000 rpm to the chuck's 2500 rpm); reflect the physics: “to tune a radio” = to select a frequency in the RF spectrum, not “to turn the knob”; boundaries drawn too wide — you include functions beyond your control; too narrow — you constrain the design (is a radio's output the electrical signal to the speakers or the sound from the speakers?). 2) Dissection / reverse engineering: take the device apart and ask what function each part performs (an exploded view of the DeWalt drill: cord, switch, motor, transmission — the same functional map as the glass box). 3) Enumeration: list all the functions; the trick is “what if X disappeared?” (without the bridge, the cars fall into the river $\rightarrow$ the function is to carry the load; without the piers, the deck itself would fall $\rightarrow$ hold up its own weight; the Quebec Bridge collapsed twice, in 1907 and 1916, 75 dead; without the connections to the roads you could not get onto it $\rightarrow$ connect the road network; without the dividers the oncoming streams would collide $\rightarrow$ separate the traffic — the George Washington Bridge spreads the directions over levels); another trick — ask about the life cycle (the functions of the juice container: contain the liquid; be filled and sealed; be poured; be reclosable after opening; withstand the temperatures; survive shipping; identify the product). 4) Function-means tree: the top level is the basic function; the levels alternate means (trapezoids) and the secondary functions (rectangles) generated by those means (a lighter: “ignite a material” $\rightarrow$ a flame vs a heated wire $\rightarrow$ different secondary functions: fuel for the flame and protection from burns vs converting electricity into heat). Caution: the function-means tree does not replace the problem statement and is not used alone — it can anchor the first solution that comes to mind (“don't marry your first design”). Objectives vs functions: objectives are adjectives (what it should be like), functions are “verb–noun” (what it should do).

Moaveni, chapter 3 “Introduction to Engineering Design” — the eight basic steps of the design process (for all engineers, regardless of background): 1) recognise the need for a product or a service; 2) define and fully understand the problem (the most important step: to fully understand the problem = to ask many questions — how much money? what size and material restrictions? when is it needed? how many units?); 3) preliminary research and preparation (do not reinvent the wheel; gather the maximum of information from the client, the engineers, the technicians, through internet searches; organise it); 4) the conceptualisation of ideas (generate ideas without a detailed analysis; do not discard reasonable concepts; for complex systems — identify the components; the question: “would the concepts work if they were developed?”; a plan of milestones — a milestone chart); 5) synthesis (consider the details: calculations, computer models, narrowing down the materials, the sizes of the components, how to manufacture; check against the codes and the standards); 6) evaluation (a detailed analysis; the critical parameters; where there is uncertainty — an experiment; working

models; choose the best solution among the alternatives); 7) optimisation (minimisation/maximisation; a functional design versus an optimised one: a functional design meets all the pre-established requirements but admits improvements; the example of a 3 m ladder designed for 1335 N with a factor of safety); 8) the presentation of the final solution (the steps are not independent; the clients change the parameters → a return to steps 1–2; oral and written reports — constantly).

Moaveni ch. 3, additional topics: economics in engineering decisions; material selection; the common traits of good teams; project management; the patent, the trademark, the copyright; why standards and codes are needed; the role of the large standardisation organisations — ANSI, ASTM, BSI, CSA, ISO, CSBTS, SIS, CEN, AFNOR, DIN; the EPA and the standards for drinking water and for outdoor and indoor air; the sources of pollutants. The end-of-chapter assignments (Problems 3.1–3.29, the key ones listed): 3.1 list 10 products that are continually modified for new technologies; 3.2 five products that are not on the market but would be useful; 3.3 five sporting goods that would be worth designing; 3.4 five internet services of the future; 3.5 research the design of bottle caps (Pepsi/Coke, aspirin, shampoo, mouthwash, ketchup) — the parameters, the pros and the cons; 3.6 research the design of clips (paper clips, hair clips, chip bag clips); 3.7 two concepts for serving the customers at peak hours in a grocery store; 3.8 two methods for an airline to deliver baggage from home to the final destination; 3.9 a flight to Mars: the schedule and the launch window, the food reserves, the exercise machines against atrophy, the waste, the energy, the key parameters; 3.10 research the design of five pens and mechanical pencils; 3.11 in-class: a boat from a 12×12 inch sheet of aluminium that holds the maximum number of coins; 3.12 in-class: a bridge of straws and paper clips between chairs 18 inches apart that holds at least 3 pounds; 3.13 ten household products certified by Underwriters Laboratories; 3.14 a table of hat sizes US/Europe; 3.15 a table of metric and U.S. wrench sizes; 3.16 the resistor colour codes (Black...Silver) with two examples of reading them; 3.17 steel pipes: the nominal size, the schedule, the inside/outside diameter, the wall thickness, the area; 3.18 the American Wire Gauge: the gauge number, the diameter in mils, the cross-section, the resistance per 1000 feet; 3.19 the role and the function of the U.S. Department of Transportation (DOT); 3.20 the highway sign fonts (Standard Alphabets for Highway Signs); 3.21 the Nuclear Regulatory Commission (NRC); 3.22 the classification of fire extinguishers: class A, B, C, D fires; 3.23 the history of seat belts; 3.24 the missions of the standardisation organisations (CENELEC, ETSI, COPANT, BID, HKSTC, KAITECH, PSB, SNZ); 3.25 the ISO 9000 and ISO 14000 certifications; 3.26 the chemical composition of the city water: ask the supplier how it is treated; the EPA hotlines; 3.27 the Surface Water Treatment Rule (SWTR) standards; 3.28 the EPA fact sheets: antimony, barium, beryllium, cadmium, cyanide, mercury; 3.29 OSHA and the 1970 Occupational Safety and Health Act (Public Law 91-596) — which standards it covers. Plus the built-in assignment “Impromptu Design II”: a tower of maximum height from an 8.5×11 inch sheet of paper and 20 inches of tape, standing for at least a minute; 30 minutes to prepare; and the “Brainstorming Session” block: define the goal and the rules of the session (do not criticise ideas while they are being offered; one person speaks; the facilitator watches the rules; record all the ideas where everyone can see them; do not be afraid to look silly; do not tie an idea to its author — the idea belongs to everyone); after the brainstorm — single out the promising ones (do not evaluate them in detail!), discuss the improvements, choose some for a detailed evaluation, evaluate them.

Moaveni, chapter 5 “Engineering Ethics” (a supplement to Week 1, directly adjacent to the theme of responsibility for engineering decisions): the full structure of the NSPE Code of Ethics (Honesty, integrity;

Fundamental Canons; Rules of Practice; Professional Obligations); the Engineer's Creed (see section 2.2); the definitions of academic dishonesty, plagiarism, conflict of interest, contract, professional responsibility; the assignments: the 25 statements of the NSPE “true/false” test (among them: engineers may perform services outside their areas of competence if they notify their employer — false; engineers may make subjective statements in a letter — false; engineers must notify their employer and other appropriate authorities if their judgement is overruled under circumstances that endanger life or property — true; engineers, knowing of a violation of the code, must report it to the professional bodies after 30 days if it is not corrected — true; engineers do not sign plans outside their competence — true; engineers do not accept compensation from more than one party for the same project without disclosure — true; engineers admit their own errors — true; and so on) and the case analyses of the NSPE Board of Ethical Review (accepting someone else's technical proposal without consent — unethical; offering engineering services on the basis of a design CD-ROM without experience — unethical: competence is not bought on a disk; a letter of recommendation without personal supervision — it is unethical to claim direct supervision).

5.4. Week-4 atamalar (terms) — glossary

Term (EN)	Atama (UZ)	One-phrase definition
Fact	Fakt	Something observed or measured, with the method and the time period
Inference	Xulosa (inference)	A conclusion read out of facts
Speculation	Taxmin (speculation)	A guess about a cause without verified evidence
Opinion	Fikr (opinion)	A personal view or preference; useful for understanding what people care about
Complaint	Shikoyat (complaint)	The symptom — what people say
Need	Ehtiyoj (need)	What the person actually requires
Problem statement	Muammo bayoni	[Stakeholder] needs [outcome] in [system], judged by [measure], within [constraints]
Solution-free test	Yechimsizlik testi	The statement must not name a device, an app, a vehicle or a building
Performance variable	Samaradorlik o'zgaruvchisi	What we judge: whether the system works
Design variable	Loyiha (design) o'zgaruvchisi	What we can change: the levers
Context variable	Kontekst o'zgaruvchisi	What we cannot change: the limits
Demand (D)	Talab (demand)	Must be met; otherwise the design is unacceptable
Wish (W)	Istak (wish)	Nice to have; not critical; do not turn it into a demand
Performance measure	Samaradorlik o'lchovi	Measurable, relevant, solution-free, specific (units + target)

Term (EN)	Atama (UZ)	One-phrase definition
Need-Know-How-Solve	Need-Know-How-Solve	Need (what we seek) — Know (what we know) — How (the method and the assumptions) — Solve (compute and check)
Design objective	Dizayn maqsadi	A property/behaviour we want; it may be achieved partially
Constraint	Cheklov	A binary limit; violating it makes the design unacceptable
Function	Funksiya	What the device must do; usually a transformation of energy/information/material
Means	Vosita (means)	A way to accomplish a function; tied to a solution
Metric	Metrika	The scale on which achievement of an objective is measured
Specification	Spetsifikatsiya	The scale on which the performance of functions is measured; the engineering requirements
Surrogate metric	O'rinbosar metrika	A measurable stand-in for a hard-to-measure objective (dropping the phone — strength)
Satisficing	Satisficing	Satisfying all the constraints without necessarily being optimal (Simon)
Objectives tree	Maqsadlar daraxti	A hierarchy of objectives: “why?” goes up, “how?” goes down
Pairwise comparison chart	Juftlik solishtirish jadvali	A matrix comparing the objectives with one another, for prioritising
Function-means tree	Funksiya-vosita daraxti	The means and the secondary functions they generate; beware of anchoring on the first solution
Black / glass box	Qora/shisha quti	Inputs–outputs without the internals; then the opened-up subfunctions
Reverse engineering	Teskari muhandislik	Taking a device apart to understand the functions of its parts
Design notebook	Dizayn daftari	A numbered, permanently bound project notebook; the basis of patent disputes
Eight-step design process	Sakkiz qadamli dizayn jarayoni	Define → generate → evaluate/select → detail → defend → manufacture/test → evaluate → report

Table 5.1. Week-4 terms (English — Uzbek)

6. Formulas (Formulas) — consolidated reference of all course formulas

This section collects all the formulas and calculation relations encountered in the four weeks: in the lectures, the labs, and the textbook chapters they refer to. Every formula is given with its variables and units; how to write units and the rules of significant figures are in section 7.

Formula / relation	Explanation	Source
value [units] $\times$ factor [units/units] = result	Kosky's fail-safe method: every quantity and factor is written with units in brackets; units cancel like fractions: $(10 \text{ [ft]}) \times (12 \text{ [in/ft]}) = 120 \text{ in}$	Kosky 2.4.2
$1 \text{ N} = (1 \text{ kg})(1 \text{ m/s}^2)$	The newton from the second law: the force that accelerates 1 kg at 1 m/s^2	Moaveni 6.2
$1 \text{ lb} = (1 \text{ slug})(1 \text{ ft/s}^2)$	British Gravitational: a slug under one pound of force accelerates at 1 ft/s^2	Moaveni 6.2
$1 \text{ slug} = 32.2 \text{ lbm}$; $1 \text{ lbm} = 0.4536 \text{ kg}$; $1 \text{ kg} = 2.2046 \text{ lbm}$	Mass-unit relations across the three systems	Moaveni Table 6.6
$1 \text{ ft} = 0.3048 \text{ m}$; $1 \text{ m} = 3.2808 \text{ ft}$; $1 \text{ in} = 2.54 \text{ cm}$	Length	Kosky p. 2; Moaveni
$1 \text{ lbf} = 4.448 \text{ N}$; $1 \text{ N} = 0.2248 \text{ lbf}$	Force	Moaveni Table 6.6
$T(^{\circ}\text{F}) = 9/5 \cdot T(^{\circ}\text{C}) + 32$; $T(^{\circ}\text{C}) = 5/9 \cdot (T(^{\circ}\text{F}) - 32)$	Celsius–Fahrenheit	Moaveni 6.2
$\text{K} = ^{\circ}\text{C} + 273.15$; $^{\circ}\text{R} = ^{\circ}\text{F} + 459.67$; $T(^{\circ}\text{R}) = 9/5 \cdot T(\text{K})$	Absolute scales	Moaveni 6.2
$1 \text{ J} = 0.7375 \text{ ft}\cdot\text{lbf}$; $1 \text{ ft}\cdot\text{lbf} = 1.3558 \text{ J}$; $1 \text{ Btu} = 778.17 \text{ ft}\cdot\text{lbf}$	Energy	Moaveni Table 6.6
$1 \text{ W} = 1 \text{ J/s} = 0.7375 \text{ ft}\cdot\text{lbf/s}$; $1 \text{ hp} = 550 \text{ ft}\cdot\text{lbf/s} = 746 \text{ W}$; $1 \text{ W} = 0.7457 \text{ kW}$	Power	Kosky p. 2; Moaveni
$V_{\text{molar}} = RT/p = 24.45 \text{ L/mol}$ (at 1 atm, 25 $^{\circ}\text{C}$)	Molar volume of an ideal gas from $pV = nRT$, $R = 8.3143 \text{ N}\cdot\text{m}/(\text{mol}\cdot\text{K})$	Kosky 18.3.3

Table 6.1. Measurement and unit formulas (Kosky ch. 2; Moaveni ch. 6)

Formula / rule	Explanation	Source
Rule 1 (addition/subtraction): the sum/difference has no significant digits to the right of the decimal position of the least precise number	$113.2 + 1.43 = 114.6$; $113.212 - 113.0 = 0.2$ (one significant figure)	Kosky 2.5

Formula / rule	Explanation	Source
Rule 2 (multiplication/division): the product/quotient carries no more significant figures than the least precise of the numbers	$(113.2) \times (1.43) = 162$; $113.2/1.43 = 79.2$	Kosky 2.5
Rule 3 (rounding): a dropped 0–4 leaves the digit unchanged; 5–9 increases it by 1	$1.54 \rightarrow 1.5$; $1.55 \rightarrow 1.6$	Kosky 2.5
Intermediate results are not rounded; the rules apply only to the final answer	Area of a circle: $R = 8.5[\text{in}]/12[\text{in}/\text{ft}]$; $A = \pi R^2 = 1.8 \text{ ft}^2$ (2 sig. figs.)	Kosky 2.5.1
$C_A = m_A/V_{\text{mix}}$	Concentration of pollutant A: mass per volume of the mixture (18.1)	Kosky 18.3
For water: 1 ppm = 1 mg/L; 1 ppb = 1 $\mu\text{g}/\text{L}$; 1 $\text{g}/\text{m}^3 = 1 \text{ mg}/\text{L} = 1 \text{ ppm}$	Aqueous concentrations	Kosky 18.3.1
For soil: 1 ppm = 1 mg/kg; 1 ppb = 1 $\mu\text{g}/\text{kg}$	Soil concentrations	Kosky 18.3.2
$\text{Conc. (mg}/\text{m}^3) = \text{Conc. (ppm)} \times M/24.45$	Gas concentration conversion; M is the molar mass, g/mol (18.2a)	Kosky 18.3.3
$\text{Conc. (ppm)} = \text{Conc. (mg}/\text{m}^3) \times 24.45/M$	Inverse conversion (18.2b)	Kosky 18.3.3
$(m)_{\text{CV}} = (m)_{\text{in}} - (m)_{\text{out}} + G$	Mass balance: accumulation = input – output + generation (18.3a)	Kosky 18.4
$(\dot{m})_{\text{CV}} = \Sigma(\dot{m})_{\text{in}} - \Sigma(\dot{m})_{\text{out}} + \dot{M} G$	Rate form of the mass balance, $\dot{m} = dm/dt$ (18.3b)	Kosky 18.4
$\dot{m}_p = C_p \cdot Q_f$	Pollutant mass flow rate = concentration $\times$ volumetric flow rate (18.4)	Kosky 18.4
$0 = \Sigma(C_p Q_f)_{\text{in}} - \Sigma(C_p Q_f)_{\text{out}} + \dot{M} G_p$	Pollutant balance with no accumulation (18.5); plus the auxiliary equation $C_{\text{carrier}} = 1.0$	Kosky 18.4
Example 18.2: $\dot{m}_{\text{LF}} = 1000 - (0.25 \times 1000 + 200) = 550 \text{ tons/day}$	Transfer Center: 600 + 400 per day, 25% recycled, 200 to the incinerator	Lecture 3; Kosky 18.4
Example 18.1: $405 \text{ ppm CO}_2 = 405 \times 44/24.45 = 729 \text{ mg}/\text{m}^3$	CO_2 concentration conversion, $M = 44 \text{ g}/\text{mol}$	Kosky 18.3

Table 6.2. Significant-figure formulas (Kosky 2.5) and environmental calculations (Kosky ch. 18)

Formula	Explanation	Exercise
Ranch cost: $4X \times 10^4 \text{ \$}/\text{mile}$? no: $4X [\text{mi}] \times 10^3 [\text{\$}/\text{mi}] + X^2 [\text{mi}^2] \times 10^4 [\text{\$}/\text{mi}^2] = \$320,000$	A square ranch with side X: perimeter 4X, area X^2 ; the answer is $X = 1.6 \text{ miles}$	Example 2.3
$X_2/X_1 = L_2/L_1$	Wire stretch is proportional to length (Hooke's law in spirit): $15.0 \times 4.00/2.00 = 30.0 \text{ mm}$	Example 2.5
Gallons = miles / mpg; total = Σ over segments	Rental-company spreadsheet: sums and averages over the segments (city/suburban/highway)	Examples 2.6–2.9

Formula	Explanation	Exercise
$\text{mph} = 136.5(1 - e^{-0.158t})$, t in seconds	The manufacturer's speed characteristic, for comparison with measurements (table versus theory)	Example 2.10
$G = W^{2/3}$; $\text{mpg} = 84500/W - 3.0$; $\text{range} = \text{gallons} \times \text{mpg}$	Weight of gasoline, fuel consumption, and driving range of a car	Ex. 17–19
$X = W/K + \sqrt{(W^2/K^2 + 2WL/K)}$	Stretch of a jumper's elastic cord: weight W (lbf), stiffness K (lbf/ft), length L (ft); in the spreadsheet: $X = W/K + \text{sqrt}(W^2/K^2 + 2*W*L/K)$	Ex. 20–23
$\text{Cost/mile} = 0.20/\text{year} \times \text{price}/(\text{miles per year}) + (\text{gasoline price}) \times (\text{gallons/mile})$	Car operation; \$5.00/gallon, 2.0×10^4 miles/year	Ex. 24–26

Table 6.3. Calculation relations from the Kosky ch. 2 exercises (models for weeks 2–4)

A method reminder for all formulas: first Need (what we are looking for and in what units), then Know (what is given and what is known from experience), then How (which equation and which assumptions), and only then Solve — with a units check in square brackets and the correct number of significant figures. Example 2.4 (barbershops): $30,000 \text{ men} \times 10 \text{ haircuts/year} = 300,000 \text{ haircuts/year} = 1,000 \text{ haircuts/day}$ (300 days) $\rightarrow 62.5 \text{ barbers}$ (16 haircuts per day) $\rightarrow 62.5/3 \approx 21 \text{ barbershops}$; the directory check gives 23, an error of about 10% — the order of magnitude checks out.

7. Qoidalar (Rules) — every course rule in one checklist

7.1. Process and diagram rules

1. The five rules of a readable PFD: one start; every path reaches an end; steps are “verb + noun”; labels on all arrows; at every decision both outputs (Yes and No) lead somewhere; one level of detail per diagram.
2. The five PFD symbols of the course: Start/End, Step, Decision, Flow, Wait/queue (dashed frame). Sequence matters more than the set of symbols.
3. Three levels of detail: Level 0 black box (inputs/outputs), Level 1 main steps, Level 2 zoom in; choose the level to fit the question.
4. Every arrow carries one of the five flow categories: materials, people, information, money, energy.
5. The system boundary is drawn, not discovered: declare it, justify it, and keep it consistent; arrows crossing the boundary = inputs and outputs.
6. A system is not a heap of parts: the parts must interact purposefully, transforming inputs into outputs.

7.2. Problem statement and variable rules

1. The solution-free test: a problem statement must not name a device, an application, a vehicle, or a building; otherwise rewrite it.
2. Statement template: [Stakeholder] needs [outcome] in [system], judged by [measure], within [constraints].
3. The path: complaint (symptom) → need (what is required) → problem statement (what we are solving).
4. Sort what you hear: fact / inference / speculation / opinion; only facts and good inferences may anchor the statement; verify what you hear: hear → observe and measure → decide.
5. A variable = a number + units; if you cannot picture a measurement, keep searching; a useful variable answers “how much?/how many?”.
6. Three kinds of variables: performance (we are judged on it), design (the levers), context (the limits); at each PFD step — one variable with its unit.
7. Specifications: each one is a D (demand) or a W (wish); do not turn a wish into a demand; numbers wherever possible, even rough ones; specifications are independent of the solution.
8. A good performance measure: measurable, relevant, solution-free, concrete (units + target); beware of averages that hide long tails.
9. Working definition of performance: [system] is judged by [variables with units and targets], through the eyes of [stakeholders]; write it down and keep the team consistent on it.
10. Need-Know-How-Solve: write Need first; Know — what is known; How — the method and the assumptions before computing; Solve — compute, check the units, interpret.

7.3. Rules of units, significant figures, and calculations

1. An engineering calculation is right in four ways: variables, number, units, significant figures.
2. Units in calculations: [in], [ft], [in/ft] — brackets on every quantity and factor; units cancel like fractions.
3. Unit names are lowercase (newton); abbreviations derived from names are capitalized (N); no plurals (kg, not kgs); no period; two letters only to distinguish units (Wb vs W) and for prefixes (kW).
4. Significant-figure rules: sum/difference — by the decimal positions of the least precise number; product/quotient — by the number of significant figures of the least precise; rounding 0–4 down, 5–9 up; do not round intermediates.
5. Check unit consistency across the whole system (the Mars Climate Orbiter lesson: pounds-force versus newtons cost the spacecraft).
6. Order-of-magnitude estimates (Fermi problems): honest rough assumptions → a corridor for the answer; a check within ~10% is already a success (21 vs 23 barbershops).

7.4. Ethical rules

1. The six NSPE canons: the safety/health/welfare of the public is paramount; practice only within your area of competence; public statements are objective and truthful; be a faithful agent or trustee; no deception; act honorably and lawfully — for the honor of the profession.
2. The extended ten (Lab 1): + no hidden payments from several parties; do not misrepresent qualifications/experience/results; give co-authors due credit; do not stay silent about violations of standards even under pressure.
3. The five cornerstones of personal ethics: do what you promised; do not disclose what was entrusted to you; accept responsibility; do not take part in lies; do not accept compromising gifts. (+ the “golden rule” as a supplement.)
4. The top question of an ethical decision: “Would I do this in front of my mother?” (the full list of 10 — in section 2.2).
5. The ethical decision matrix: rows — the canons, columns — courses of action; “yes/no/maybe” with a justification; sum the yes/no (ignore the maybes); one overriding criterion cancels the count.
6. TUT course rules: attendance $\geq 75\%$ to be admitted to the final; plagiarism: 1st offense — 0 and a possible resit, 2nd — 0 without a resit, 3rd — course failure; AI assistance is allowed with explicit citation of the tool and a description of its contribution; submitting AI-generated text as your own is an academic violation.

7.5. Design and team rules

1. A design problem is open-ended and ill-structured: there is no single answer; “best” is determined by the client's critical requirements.
2. Define the problem in terms of needs, not solutions: a solution in the statement closes the door on other solutions.
3. The eight steps of the design process (Kosky): define the problem → generate alternatives → evaluate and select → detail → defend → build and test → evaluate performance → final report.
4. The two goals of a systematic approach: remove personal biases; maximum thinking and information gathering before the final choice.
5. Design notebook: number and date the pages; do not tear pages out; do not rewrite a fair copy; use pen; strike through, do not erase; paste in the printouts; write for another reader; record the meetings.
6. Brainstorming rules: no criticism while ideas are being offered; one person speaks; a facilitator; write everything down; do not tie an idea to a person; afterwards — shortlist the promising ideas without evaluating them in detail.
7. Team versus group: a shared goal and coordination; the leader delegates, communicates, keeps humor and confidence, and sets an example.
8. Project management 3S (Dym): scope, schedule, spending; tools — Gantt (task bars over time) and PERT (a network, the critical path).

9. The client's initial statement: look for errors, biases, and implied solutions; sort the attributes into O/C/F/M; cross out the means for the time being.
10. Functions via the black box: list the inputs/outputs, then “lift the lid”; listing tricks: “what if X disappeared?” and “how do you maintain it over the life cycle?”.

7.6. Rules of the course project milestones

1. Milestone 1 (weeks 2–3): the chosen system + boundary (at least 3 inputs, 3 outputs) + a stakeholder map of Uses/Runs/Affected; then the PFD and the stakeholder map — the submittable Milestone 1.
2. Milestone 2 (week 5): a problem statement document + an explicit list of modeling assumptions.
3. Milestone 3 (week 7): the first quantitative pass of the model + a short pitch in class.
4. Milestone 4 (week 10): a trade-off analysis of at least two alternative designs.
5. Milestone 5 (week 12): a draft scenario simulation and a comparison of outcomes.
6. Labs are graded with a live checklist: Lab 2 — all six systems steps; Lab 4 — all three parts of the template (A: classification and framing; B: process and variables; C: stakeholders and goals).

8. Topshiriqlar (Assignments) — all course assignments, weeks 1–4

From the syllabus (section 7, Independent Study Assignments): independent work is structured around a continuous team project; the group tasks (at least one third of the assignments, by university convention) support the team's one system/project; the individual ones — reflection, preparation, and peer assessment. Each week has 4 hours of independent work allotted to it.

Week	Assignment	Type	Deadline
1	Individual reflection: identify three everyday systems, describe their boundaries and stakeholders — preparation for choosing the team system (announced at the end of lecture 1: a laundromat, a bus route, a cafeteria, a payment app)	individual	before lecture 2
2	Team task: finalize the choice of system; sketch the boundary and the stakeholder map (draft Milestone 1; announced at lecture 2: at least 3 inputs and 3 outputs)	team	before lecture 3
2	Lab 2 “Applied Systems Challenge”: six steps on the conflict card (system name, boundary, inputs/outputs, stakeholders, decision, context)	team, graded in class	at the lab
3	Team deliverable: a full process flow diagram and stakeholder map — submit (Milestone 1)	team	submission

Week	Assignment	Type	Deadline
3	After lecture 3 (for yourself): Level 0 and Level 1 (5–8 steps) of your own system; label the arrows; redraw a fragment from a second stakeholder's perspective; ask each stakeholder what “works well” means	not submitted	by week 4
4	Team task: a draft problem statement and a working definition of “system performance” (after lecture 4: a statement in terms of needs; 3–5 variables with units; D/W with numerical targets; the solution-free test)	team/individual	by week 5
4	Lab 4 “Applied Problem Framing”: three parts of the template — classification and framing; a process of up to 6–7 steps with a variable per step; stakeholders and goals (3+ rows, D and W with numbers)	team, graded in class	at the lab
5	Team deliverable: a problem statement document and an explicit list of modeling assumptions (Milestone 2)	team	submission

Table 8.1. Independent study assignments, weeks 1–4 (syllabus) + those issued in class

The graded work of the first weeks is the lab challenges (Lab 2: six steps; Lab 4: three parts) and the weekly mini-projects (report + presentation, 30% of the final grade). The ungraded reflections of weeks 3 and 4 are the team's working material: they turn directly into Milestone 2. Remember the attendance rule: 75% of classes is a mandatory condition for admission to the final exam.

9. Cheat sheet: key facts, numbers and quotes

9.1. Facts and numbers they ask about

Fact	Value
Course code / ECTS / semester	CC-INEN010104 · 4 ECTS · Fall 2026/27, compulsory
Workload	120 h = 30 lectures + 30 project work + 60 independent study
Grading	mini-projects 30% + midterm 30% + final 40%
Scale	A 90–100 · B 75–89 · C 60–74 · D 0–59
Attendance	at least 75% of classes to be admitted to the final
Plagiarism	1st — 0 (a resit is possible) · 2nd — 0 without a resit · 3rd — course failure
Engineers in the USA	more than 1.6 million; the second-largest profession
Starting salary (USA, 2017)	≈ \$68,000 per year
Employment by Moaveni	civil 17.7% · mechanical 15.2% · industrial 13.7% · electrical 10.0%

Fact	Value
NAE Grand Challenges	14 challenges; 6 shown in the lecture (water, infrastructure, health informatics, solar energy, cybersecurity, medicines)
Professional societies	ASME · IEEE · ASCE · AIChE · AIAA · ANS (+ ASTM, BMES per Kosky)
Mars Climate Orbiter	launched 11.12.1998, lost 23.09.1999: thrust not converted from lbf to N; atmospheric entry at 57 km instead of 140–150 km
Hyatt Regency, Kansas City, 1981	walkway collapse: the second-floor walkway hung from the fourth; 114 killed; the lesson — complete and unambiguous manufacturing specifications
Example 18.2 (Transfer Center)	600 + 400 t/day; 25% recycled; 200 t/day to the incinerator → 550 t/day to the landfill
Example 2.4 (barbershops)	an estimate of 21 barbershops vs 23 in the directory ($\approx 10\%$)
Example 2.3 (ranch)	\$320,000; fence $\$10 \times 10^3/\text{mile}$; land $\$10 \times 10^4/\text{mi}^2 \rightarrow$ side 1.6 miles
CO ₂ in the atmosphere (2017)	405 ppm = 729 mg/m ³ (M = 44 g/mol)
V _{molar} of an ideal gas	24.45 L/mol at 1 atm and 25 °C
Molar gas constant R	8.3143 N·m/(mol·K) = 8314.3 J/(kmol·K)
g (standard)	9.807 m/s ² = 32.174 ft/s ²

Table 9.1. Key numbers of the course

9.2. Quotes that embellish an answer

- “Science can amuse and fascinate us all, but it is engineering that changes the world” — Isaac Asimov.
- “Scientists investigate that which already is; engineers create that which has never been” — Albert Einstein.
- “The engineer has been, and is, a maker of history” — James Kip Finch.
- “Engineers operate at the interface between science and society” — Gordon Stanley Brown (MIT).
- “Design is not just what it looks like and feels like. Design is how it works” — Steve Jobs.
- “Great things in business are never done by one person. They're done by a team of people” — Steve Jobs (the epigraph to Kosky ch. 22).
- “Design is the activity of intending to produce a description of an artifact in terms of its organization and functioning — its interface between the inner and outer environments” — Herbert Simon (as cited in Dym).

9.3. Ten one-line answers (for quick revision)

1. An engineer = a creative person solving practical problems (ingenium).
2. Science explains, design invents the form, engineering builds what works.
3. A system = elements + processes transforming inputs into outputs, in an operating context.
4. The boundary is the engineer's choice, not a fact about the world.
5. A stakeholder = uses / runs / affected — even outside the boundary.
6. PFD = steps in order + arrows carrying what moves; 5 symbols, 5 rules, 3 levels.
7. A problem is stated as a need, not a solution (the solution-free test).
8. A variable = a number + units; performance / design / context.
9. D — a must, W — nice to have; numbers even if only rough.
10. Need → Know → How → Solve; units in brackets; significant figures only in the final answer.

9.4. What comes next in the syllabus (weeks 5–8, a preview)

Week 5 — problem statement and modeling II: engineering models as simplified representations; assumptions and their consequences (Milestone 2). Week 6 — quantitative reasoning I: demand and capacity, rates and flows, order-of-magnitude estimates (an individual problem set). Week 7 — quantitative reasoning II: reasoning from incomplete data; comparing readings of the same problem; the first quantitative pass of the model and a pitch (Milestone 3). Week 8 — Midterm: a 1–2 hour written exam, 30% of the grade, covering ILO 1–4 — that is, all the material of this guide.